Additive Manufacturing 119 (2026) 105098

Contents lists available at ScienceDirect

Additive Manufacturing

journal homepage: www.elsevier.com/locate/addma

Research paper

Substepped and advected subdomain methods for part-scale LPBF modeling

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ARTICLE INFO

Keywords:
Part-scale LPBF modeling
Substepping
Multi-timesteps
Temporal multiscale
Domain decomposition

ABSTRACT

This work presents substepping schemes for part-scale Laser Powder Bed Fusion (LPBF) modeling where the computational domain is divided into a fast partition with a small time-step and a slow one using a large time-step. These schemes are of special interest due to the disparity of time scales present in LPBF applications, which renders simulation prohibitively expensive unless simplifications are made. A Robin–Robin substepper is proposed and compared to the scheme proposed by Hodge (2021).

Additionally, the introduction of an advected subdomain in the fast partition is proposed. This approach was first presented in Slimani et al. (2024) and allows for larger time-steps, as the problem in the Heat Affected Zone (HAZ) is posed in the moving reference frame of the laser, where the thermal field is significantly steadier than in the fixed frame.

These methods are used in combination to obtain a computationally efficient scheme for LPBF thermal simulations. The proposed methods are implemented in `FEniCSx`. Their strengths and weaknesses are evaluated through 2D benchmarks, comparison with experimental melt pool data, and 3D LPBF simulations.

1. Introduction

Thermal modeling of PBF-LB/M, commonly referred to as Laser Powder Bed Fusion (LPBF), is essential for optimizing the microstructure and the scan path in order to reduce warpage and residual stresses. However, part-scale simulation is especially costly due to the "extremely multiscale" nature of the process [1].

As an example, consider the bridge structures proposed in the AM Bench 2022 challenge [2]: the smallest and largest lengths are 40 μm (laser radius) and 75 mm (part length), resulting in a ratio of spatial scales in the order of 10^3. The disparity is even greater for temporal scales: the cumulative laser-on time, T_{print}, is about 15 min, while the time it takes for the laser to move by one radius, T_{hs}, is about 40 μs, yielding a ratio on the order of 10^7.

More generally, assuming a cubic part of side length L, where both the layer thickness and the hatch spacing are on the order of the laser radius R, the ratios scale as

$$\frac{T_{print}}{T_{hs}} = \mathcal{O}\left(\frac{L^3}{R^3}\right) \tag{1}$$

imposing a prohibitively high number of time-steps to accurately resolve the Heat Affected Zone (HAZ). This issue is hard to circumvent, since, when modeling in the laboratory reference frame, if the time-step Δt does not satisfy the constraint

$$\Delta t \leq T_{hs} \tag{2}$$

the resulting temperature field exhibits spurious oscillations, rendering the solution qualitatively and quantitatively unacceptable. To achieve accurate results, typical values of the employed time-step are around 0.2–0.5 T_{hs} [3–5].

Since employing an accurate power density profile comes at a high cost, the most common approach is to utilize *lumped heat sources*, encompassing techniques such as *hatch lumping*, *layer lumping* and *flash heating* [6–8]. A time-step is decided without regard for constraint (2) and the laser's energy input is evenly distributed along its path. Although this approach allows for feasible simulations, it comes at the cost of accuracy. Van Elsen et al. [9] shows that while temperature contours far from the HAZ might coincide, peak temperature is grossly underestimated, possibly resulting in the absence of a melt pool. *Flash heating* addresses this issue by artificially increasing the deposition speed in order to recover a more faithful melt volume [4], hence requiring numerical calibration.

Recent literature has pursued two avenues to render part-scale simulations feasible without lumped heat sources: optimizing the computation of a single time-step and using small time-steps only where necessary.

Many approaches exist to reduce the cost of a given time-step, such as matrix-free FEM [10], the use of GPUs [10,11] and CPU vectorization [5]. Essongue et al. [12] advocates for explicit time integration schemes, as in LBPF applications, constraint (2) is more restrictive than

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https://doi.org/10.1016/j.addma.2026.105098
Received 19 September 2025; Received in revised form 9 December 2025; Accepted 26 January 2026
Available online 9 February 2026
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the stability threshold. In their follow-up work, Essongue et al. [13] demonstrate that the speed-up is significantly lower in Directed Energy Deposition modeling due to the reduced contrast in time scales.

Note that although Proell et al. [5] achieves a considerable speed-up by combining many of these methods together with additional compiler level optimizations, they still consider the simulation of decimeter-scale parts unrealistic.

Since constraint (2) must only be enforced near the HAZ, multiple authors have proposed methods that allow different time-steps in different subdomains. Kopp et al. [14] shows welding examples with space–time FEM: because the time axis is included in the mesh, adaptive mesh refinement is applied in order to use smaller time-steps in the HAZ. Few references have adopted this technique, since the dominant trend in the field is implementations based on finite difference discretizations in time.

Conversely, substepping has seen broader adoption due to its easier integration into existing software. The first substepping work in AM was introduced by Soldner and Mergheim [8], who combined it with a lumped heat source model. Following the approach of Nakshatrala et al. [15], they evaluated the interface terms explicitly at t^n, introducing a stability constraint on the macro time-step Δt_s.

Viguerie et al. [3] presents a substepping approach based on a custom domain decomposition (DD) scheme. In their welding examples with IN625, they employ micro-steps of $1T_{hs}$ and macro-steps of $4T_{hs}$, obtaining a speed-up of 2.5. In comparison, Hodge [1] claims much larger speed-ups in the order of 100, albeit with much smaller micro-steps of approximately $0.02T_{hs}$, and similarly sized macro-steps of $3T_{hs}$. Although the two substepping schemes are structurally similar, the treatment of the phase change in the latter necessitated very fine time discretization (apparent heat capacity and heat integration methods). In their follow-up work, Puso and Hodge [16] showcase the substepping scheme in a practical example using micro-steps of about $0.04T_{hs}$ and macro-steps of $7.7T_{hs}$.

Cheng and Wagner [17] derive an efficient substepping scheme based on Dirichlet–Robin interface conditions. The Robin coefficient is analytically derived from a linear 1D problem, resulting in high accuracy at the interface. They showcase their scheme with micro-steps of $0.07T_{hs}$ and macro-steps of $7\ T_{hs}$, yielding a reported speed-up of 100.

In their recent work, Elahi et al. [18] extend the three level scheme DD of Leonor and Wagner [11] to employ different time-steps in each level. They present an extremely efficient GPU-based scheme, using micro-steps of $0.086T_{hs}$ at the finest level, and macro-steps of $5.5T_{hs}$ at the coarsest level. It is apparent from this quick overview of the literature that most substepping references adopt macro-steps of a few T_{hs}.

In previous work, the authors pursued a different strategy to speed-up LPBF simulations by attaching a moving subdomain to the heat source and performing computations in its reference frame [19]. In this frame, where the laser remains stationary, the thermal field is steadier, allowing for significantly larger time-steps for long scan tracks. However, the work only addressed the linear heat equation and the overall speed-up was limited by the increased computational cost introduced by the advection terms.

Although originally referred to as *Chimera* due to the use of two meshes, the core feature of the method is the subdomain attached to the heat source. The heat equation is posed in the moving reference frame, which introduces advection terms that are central to the model's behavior and performance. Hence, the name *advected subdomain* is used in this work to better reflect the method's defining characteristics.

In this work, the authors implement the substepping scheme described by Hodge [1] as well as an additional Robin–Robin scheme and add their advected subdomain method in the fast subdomain. Thermally dependent material properties as well as phase change are considered in order to present a realistic thermal model of LPBF. The numerical examples will examine both the accuracy and efficiency of the various models proposed.

2. Methodology

This section describes the modeling frameworks adopted in this work. First, the balance of energy equation is presented, along with the phase change and growing domain treatments. Next, a Robin–Robin variant of the advected subdomain method presented in [19] is described. The section concludes with a short overview of multi-timestep methods and details the substepping approaches used in this work. Specifically, the substepper presented in Hodge [1] is used throughout the work as a baseline, and a novel Robin–Robin substepper with Robin coefficients based on Chen et al. [20] is proposed.

2.1. Problem statement

The governing PDE is the energy balance including diffusion, heating and latent heat effects; a temperature field $T = f_{(\mathbf{x},t)} : \Omega \to \mathbb{R}$ is sought such that

$$\begin{aligned} \rho c_p \partial_t T_{(\mathbf{x},t)} + \rho L \partial_t f_l - k\Delta T_{(\mathbf{x},t)} &= r_{(\mathbf{x},t)} && \forall \mathbf{x} \in \Omega(t) \\ -k\partial_n T &= h(T - T_{\text{env}}) + \varepsilon\sigma(T^4 - T_{\text{env}}^4) && \forall \mathbf{x} \in \partial\Omega(t) \\ T_{(\mathbf{x},0)} &= T_{0(\mathbf{x})} && \forall \mathbf{x} \in \Omega(0) \end{aligned} \tag{3}$$

where ρ, c_p, L, f_l and r denote the density, specific and latent heats, liquid fraction and power density, while h_{conv}, ε, σ, T_{env} and T_0 represent the convection heat transfer coefficient, emissivity, Stefan–Boltzmann constant and the environment and initial temperatures, respectively. Fig. 1 presents a schematic of the computational domain.

This work adopts the latent heat treatment proposed in Kollmannsberger et al. [21], which uses a variant of the source-based approach presented in Celentano et al. [22]. Rather than using the chain rule to express $\partial_t f_l$ as a function of $\partial_t T$, as done in the apparent heat capacity method, the time derivative is kept as-is [22] and f_l is regularized using a hyperbolic tangent [21]:

$$f_l(T) = \frac{1}{2}\left[\tanh\left(S\frac{2}{T_l - T_s}\left(T - \frac{T_s + T_l}{2}\right)\right) + 1\right] \tag{4}$$

where T_l, T_s and S correspond to the liquidus and solidus temperatures and a smoothing parameter. f_l and its derivative are plotted in Fig. B.25 for several values of S: as S increases, f_l approaches the step function and the phase change becomes isothermal. $\partial_t f_l$ is computed as part of the time integration scheme and can be regarded as an additional heat sink depending on the temperature history. The authors find that this approach is much more robust than the apparent heat capacity method, see Appendix B for a comparison.

The time dependency of the computational domain in Eq. (3) arises from the domain growth associated with the LPBF recoating process. It is numerically modeled using the born-dead element technique [23]. The deposition and melting processes are illustrated in fig. 2. Fig. 2(a) shows a bare substrate below an *inactive* powder layer, which does not take part in the thermal problem. Fig. 2(b) shows the activation of the powder layer, which will be subsequently heated by the moving heat source. Then, at the end of a given time-step, powder elements whose average temperature surpasses the melting point $T_m = \frac{T_s+T_l}{2}$ are set to bulk, as shown in Fig. 2(c). Following Kollmannsberger et al. [21], the material properties of both powder and bulk are temperature-dependent and become identical above the melting point, so that powder elements are indistinguishable from bulk elements once melted.

In the remainder of this work, the solution of Eq. (3) using FEM and uniform Backward-Euler time integration is referred to as the *standard model*, and serves as a reference for the proposed substepping schemes.

2.2. Advected subdomain model

The motivation for the advected subdomain method lies in the observation that the strong transience of the HAZ is primarily due to the motion of the heat source. By reformulating the problem in the frame

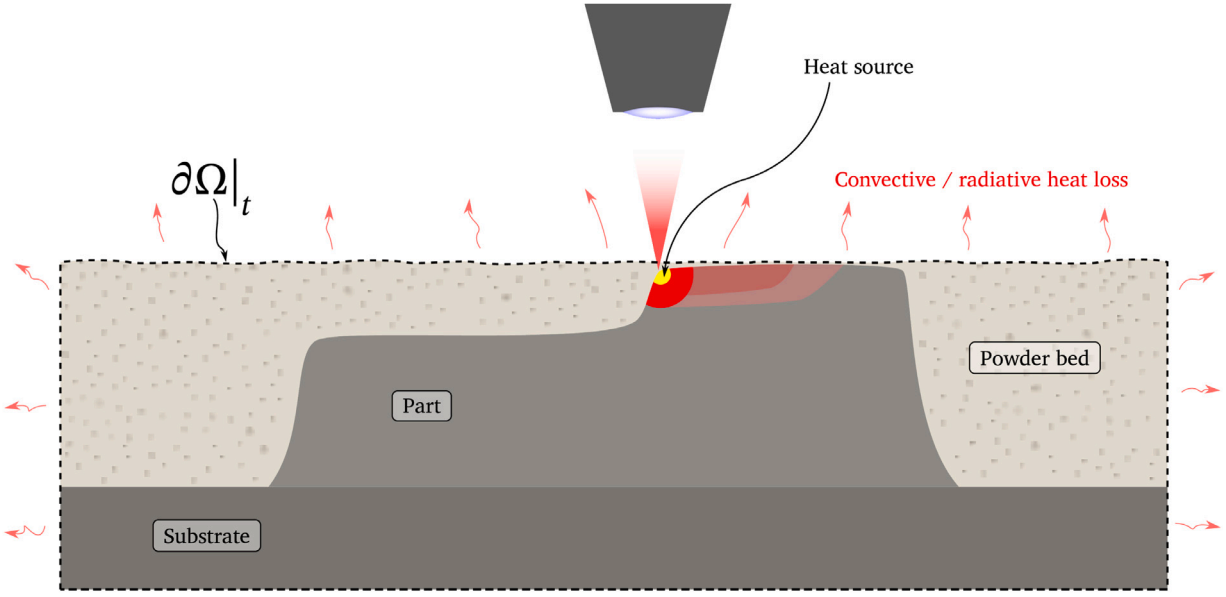
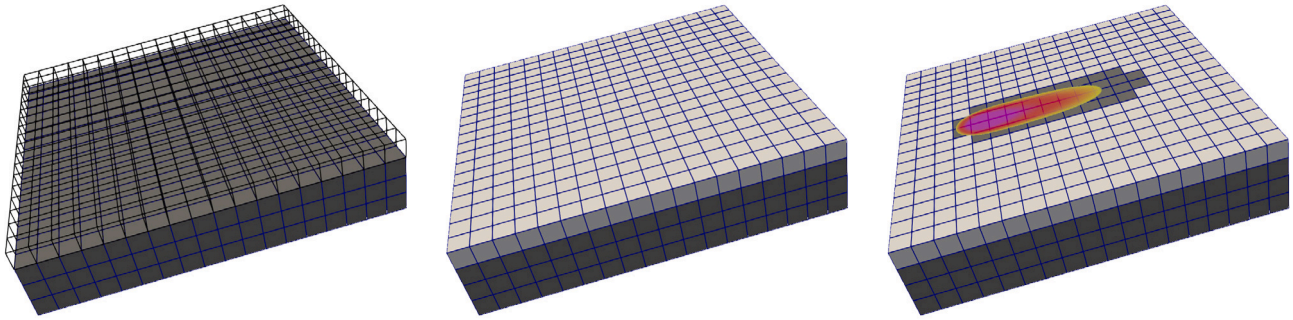


Fig. 1. Schematic of the LPBF computational domain $\Omega(t)$, encompassing the bulk (part and substrate) and powder bed regions, together with the applied heat source and convective/radiative heat losses.



(a) Bare substrate below an inactive powder layer.

(b) A powder layer is activated during a recoating step.

(c) After a heating step, elements whose average temperature surpasses T_m are set to bulk.

Fig. 2. Illustration of deposition and melting processes. ( ) Powder ( ) Bulk.

of the moving heat source, the thermal field becomes significantly steadier, allowing for much larger time-steps.

However, this approach is only meaningful in the vicinity of the heat source, as it would introduce unnecessary complexity elsewhere. Hence, in their previous work [19], the authors subdivided the computational domain in two zones: a moving subdomain $\Omega_m(t)$ attached to the heat source, where the governing equations are posed in the moving reference frame, and a fixed subdomain $\Omega_f(t) = \text{int}\big(\Omega \setminus \Omega_m(t)\big)$, where the equations remain unaltered. Let $\mathbf{v} = \frac{d\mathbf{x}_{hs}}{dt}$ denote the laser velocity; the change of coordinate system can be expressed as

$$(\mathbf{x}, t) \longrightarrow (\boldsymbol{\xi}, \eta), \begin{cases} \boldsymbol{\xi} = \mathbf{x} - \int_0^t \mathbf{v}(t)dt \\ \eta = t \end{cases}$$

$$\Longrightarrow \begin{cases} \partial_{x_i} & = \partial_{\xi_i} \Rightarrow \nabla_{\mathbf{x}} = \nabla_{\boldsymbol{\xi}} \\ \partial_t & = \partial_\eta - v_i \partial_{\xi_i} = \partial_\eta - \mathbf{v}(t) \cdot \nabla_{\boldsymbol{\xi}} \end{cases}$$

Denoting η by t for clarity, Eq. (3) rewrites as follows for $T_m : \Omega_m \to \mathbb{R}$:

$$\rho c_p \partial_t T_{m(\boldsymbol{\xi},t)} + \rho L \partial_t f_l - \rho(c_p + L f_l')\mathbf{v} \cdot \nabla_{\boldsymbol{\xi}} T_{m(\boldsymbol{\xi},t)} - k\Delta_{\boldsymbol{\xi}} T_{m(\boldsymbol{\xi},t)} = r_{(\boldsymbol{\xi})} \qquad \forall \boldsymbol{\xi} \in \Omega_m(t) \tag{5}$$

where the source term only depends on space since it is stationary in the moving frame.

As mentioned earlier, the PDE on Ω_f remains unchanged with respect to Eq. (3). If the support of the source term is fully contained in Ω_m, the right-hand side is null:

$$\rho c_p \partial_t T_{f(\mathbf{x},t)} + \rho L \partial_t f_l - k\Delta T_{f(\mathbf{x},t)} = 0 \qquad \forall \mathbf{x} \in \Omega_f(t) \tag{6}$$

In [19], the authors considered Dirichlet–Neumann coupling between the two subdomains. In this work, Robin–Robin coupling is adopted instead, as it is arguably easier to implement (same type of condition on both sides of the interface) and exhibits better staggered convergence properties than Dirichlet–Neumann [24]. Fig. 3 illustrates a non-overlapping decomposition of Ω into Ω_m and Ω_f and the Robin–Robin coupling conditions applied on the interface $\Gamma(t) = \partial\Omega_m \cap \partial\Omega_f$:

$$T_m + k\Big|_{T_m^{n+1}} \partial_{n_m} T_m = T_f + k\Big|_{T_f^{n+1}} \partial_{n_m} T_f \qquad \boldsymbol{\xi} \in \Gamma(t) \tag{7}$$

$$T_f + k\Big|_{T_f^{n+1}} \partial_{n_f} T_f = T_m + k\Big|_{T_m^{n+1}} \partial_{n_f} T_m \qquad \mathbf{x} \in \Gamma(t) \tag{8}$$

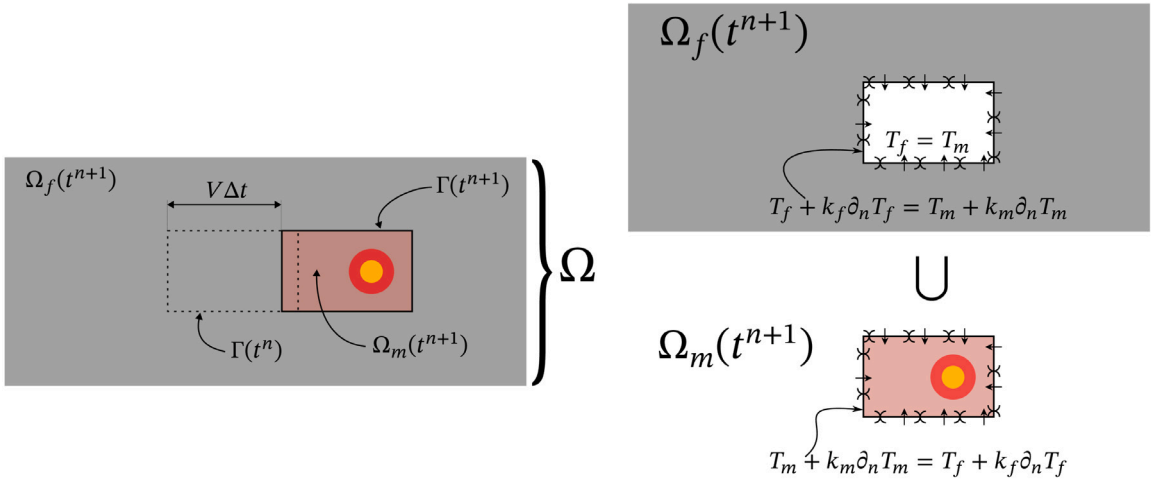


Fig. 3. Schematic of non-overlapping advected subdomain with Robin–Robin coupling. The arrows represent fluxes, and the links embody coupling of the solutions on either side. The combination of arrows and links represents the Robin conditions.

where ∂_{n_i} denotes the outward normal derivative on $\partial\Omega_i$. The remaining heat transfer and initial conditions take the form

$$
\begin{aligned}
-k\partial_n T_m &= h_{\text{conv}}\left(T_{m(\xi,t)} - T_{\text{env}}\right) + \varepsilon\sigma\left(T_{m(\xi,t)}{}^4 - T_{\text{env}}^4\right) && \xi \in \partial\Omega_m(t) \setminus \Gamma(t)\\
T_{m(\xi,0)} &= T_{0(\xi,0)} && \xi \in \overline{\Omega}_m(0)\\
-k\partial_n T_f &= h_{\text{conv}}\left(T_{f(\mathbf{x},t)} - T_{\text{env}}\right) + \varepsilon\sigma\left(T_{f(\mathbf{x},t)}{}^4 - T_{\text{env}}^4\right) && \mathbf{x} \in \partial\Omega_f(t) \setminus \Gamma(t)\\
T_{f(\mathbf{x},0)} &= T_{0(\mathbf{x},0)} && \mathbf{x} \in \overline{\Omega}_f(0)
\end{aligned}
$$

Let $\mathcal{T}_m(t)$ and $\mathcal{T}_f(t)$ denote the active finite element meshes of Ω_m and Ω_f. Let V_m and V_f be the finite element spaces associated with $\mathcal{T}_m$ and $\mathcal{T}_f$. The residual weak forms $\mathcal{R}_m$ and $\mathcal{R}_f$ of the moving and fixed subproblems are included in Appendix A. Together, they define a coupled system of nonlinear integral equations for $T_m^{n+1} \in V_m$ and $T_f^{n+1} \in V_f$:

$$
\begin{aligned}
\mathcal{R}_m(T_m^{n+1}, T_f^{n+1}) &= 0 && \forall \phi_m \in V_m\\
\mathcal{R}_f(T_m^{n+1}, T_f^{n+1}) &= 0 && \forall \phi_f \in V_f
\end{aligned}
$$

These equations are solved using Newton–Raphson iterations: given current guesses $T_m^{n+1,i}$ and $T_f^{n+1,i}$, find δT_m and δT_f such that

$$\frac{\partial \mathcal{R}_m}{\partial T_m}\delta T_m + \frac{\partial \mathcal{R}_m}{\partial T_f}\delta T_f = -\mathcal{R}_m \qquad \forall \phi_m \in V_m \tag{9}$$

$$\frac{\partial \mathcal{R}_f}{\partial T_m}\delta T_m + \frac{\partial \mathcal{R}_f}{\partial T_f}\delta T_f = -\mathcal{R}_f \qquad \forall \phi_f \in V_f \tag{10}$$

and update the solutions $T_m^{n+1,i+1} \leftarrow T_m^{n+1,i} + \delta T_m$ and $T_f^{n+1,i+1} \leftarrow T_f^{n+1,i} + \delta T_f$ until the increments δT_m and δT_f are sufficiently small.

If material properties are averaged over each element, the matrix forms of $\frac{\partial \mathcal{R}_f}{\partial T_f}$, $\frac{\partial \mathcal{R}_f}{\partial T_m}$ and $\frac{\partial \mathcal{R}_m}{\partial T_f}$ are symmetric. In contrast, the Jacobian block $\frac{\partial \mathcal{R}_m}{\partial T_m}$ is non-symmetric due to advection. When both subproblems are solved together i.e. monolithically, the overall system matrix is therefore non-symmetric. Alternatively, a staggered approach can be employed in which the moving and fixed subproblems are solved sequentially, feeding the solution of one into the other until convergence. This confines the asymmetry to the moving subproblem, but requires an additional loop on top of the Newton–Raphson iterations to enforce coupling consistency.

Both monolithic and staggered approaches are implemented in this work, and their performance is compared in Section 4.3.

As mentioned earlier, the goal of the advected subdomain method is to enable larger time-steps as the thermal field in Ω_m becomes steadier. In [19], the authors measured the steadiness of the solution by computing the relative L2 norm of the increment between two consecutive time-steps. In this work, the maximum value of the time derivative of T_m is adopted instead. When this value falls below a threshold ϵ, the time-step size Δt is increased for the next step:

$$
\begin{aligned}
&\texttt{steadiness_metric}(T_m) := \max_{\Omega_m}\left|\frac{T_m^{n+1} - T_m^n}{\Delta t}\right| < \epsilon\\
&\quad\Longrightarrow \quad \text{Increase } \Delta t \text{ for } t^{n+2} = t^{n+1} + \Delta t
\end{aligned}
\tag{11}
$$

The maximum is more robust than the relative L2 norm, as the latter is sensitive to the size of Ω_m.

The motion of Ω_m requires a special workflow of boolean operations to ensure that $\Omega_m(t^n)$ and $\Omega_m(t^{n+1})$ remain subsets of Ω at all times. Mesh intersection and subtraction are implemented efficiently using bounding box trees, making their computational cost negligible compared to assembly and solve. Additional details on the workflow itself can be found in [19], or in the open-source code repository [25].

2.3. Substepping

Substepping methods are known under various names in the literature, including multirate, subcycling, local time-stepping, multistep, etc. They consist in partitioning the domain into subdomains and marching them in time with their own time-steps and/or integration schemes.

Fig. 4 illustrates a disjoint partition $\Omega = \Omega_f \sqcup \Omega_s$, where the subscripts f and s denote the *fast* and *slow* subdomains, respectively. Let $\Gamma_{fs} = \Gamma_{sf}$ denote the interface between the two partitions. The slow partition Ω_s advances from t^n to $t^{n+1} = t^n + \Delta t_s$ in a single *macro-step*, while the fast partition Ω_f subdivides the same interval in M *micro-steps* of size $\Delta t_f = \frac{\Delta t_s}{M}$.

Let $T_s : \Omega_s \to \mathbb{R}$ and $T_f : \Omega_f \to \mathbb{R}$ denote the solutions in the slow and fast partitions. Following domain decomposition principles, they must be coupled across the interface to ensure that *transmission conditions* are satisfied. Let the superscript $\frac{n}{M}$ denote the time index at the n-th micro-step, $t^n + m\Delta t_f$, $m \in [1, M]$. The transmission conditions read

$$T_f^{n+\frac{m}{M}} = T_s^{n+\frac{m}{M}} \qquad k\partial_n T_f^{n+\frac{m}{M}} = k\partial_n T_s^{n+\frac{m}{M}} \tag{12}$$

$$T_s^{n+\frac{m}{M}} \approx (1 - \frac{m}{M})T_s^n + \frac{m}{M}T_s^{n+1} \tag{13}$$

$m \in [1\,..\,M]$

That is, both the solution and its flux must remain continuous across the interface throughout the substepping cycle. In a FEM setting, flux

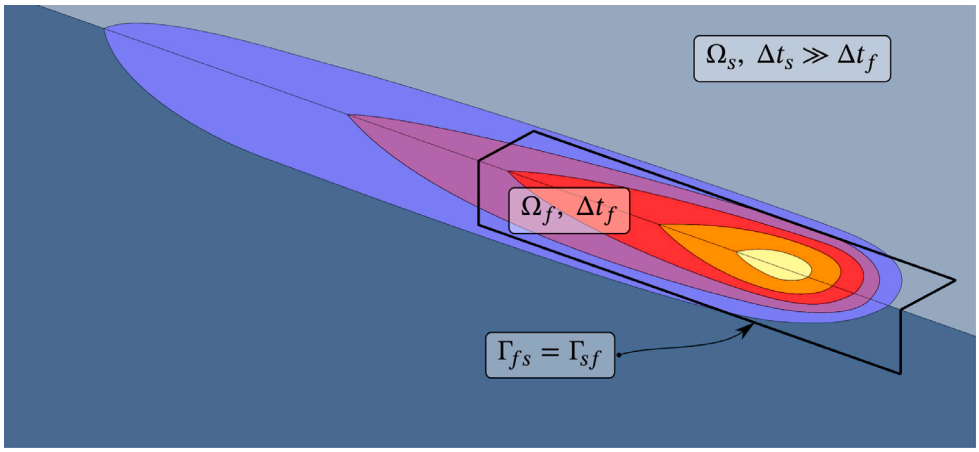


Fig. 4. Non-overlapping substepping decomposition. The HAZ is included in the fast partition Ω_f. The remainder of the domain belongs to the slow partition Ω_s.

continuity is satisfied only in the weak sense via the variational formulation. Since the slow solution is only available at macro-step endpoints, it is approximated at the micro-steps using linear interpolation (13), which introduces a consistency error of order $\mathcal{O}(\Delta t_s)$. The impact of this approximation on the overall accuracy of the substepping scheme is revisited in Section 4.1.2.

A review of the literature reveals several substepping schemes proposed for AM [1,3,8,17,18], and broader fields [24,26]. While these approaches differ primarily in the way the subdomains are coupled, reflecting the wide variety of domain decomposition methods available, a recurring distinction emerges: some schemes fully separate the fast and slow solves, while others perform combined solves, generally at the end of the substepping cycle. In this work, the former is referred to as *fully staggered*, while the latter is termed *semi-monolithic*.

These should be viewed from a domain decomposition perspective. In classical DD methods with two subdomains, there exists only two options to solve the coupled problem: either a staggered approach or a monolithic one. However, with three or more subdomains, intermediate strategies can be devised. Similarly, regarding the intermediate states of the fast partition as separate subdomains, the same classification applies to substepping schemes.

From this viewpoint, the *fully monolithic* approach solves all micro-steps and the macro-step simultaneously within a single global system, as done for instance in [26]. For a linear problem with Backward-Euler time integration and weak enforcement of the transmission conditions (12), the corresponding linear system can be written as:

$$\begin{pmatrix} +\frac{\mathbf{M}_f}{\Delta t_f}+\mathbf{A}_f & 0 & \dots & 0 & 0 & \frac{1}{M}\mathbf{C}_{fs} \\ -\frac{\mathbf{M}_f}{\Delta t_f} & +\frac{\mathbf{M}_f}{\Delta t_f}+\mathbf{A}_f & \cdots & 0 & 0 & \frac{2}{M}\mathbf{C}_{fs} \\ & & \ddots & & & \\ 0 & 0 & \dots & +\frac{\mathbf{M}_f}{\Delta t_f}+\mathbf{A}_f & 0 & \frac{M-1}{M}\mathbf{C}_{fs} \\ 0 & 0 & \dots & -\frac{\mathbf{M}_f}{\Delta t_f} & +\frac{\mathbf{M}_f}{\Delta t_f}+\mathbf{A}_f & \mathbf{C}_{fs} \\ 0 & 0 & \dots & 0 & \mathbf{C}_{sf} & \frac{\mathbf{M}_s}{\Delta t_s}+\mathbf{A}_s \end{pmatrix} \times \begin{pmatrix} \boldsymbol{T}_f^{n+\frac{1}{M}} \\ \boldsymbol{T}_f^{n+\frac{2}{M}} \\ \vdots \\ \boldsymbol{T}_f^{n+\frac{M-1}{M}} \\ \boldsymbol{T}_f^{n+1} \\ \boldsymbol{T}_s^{n+1} \end{pmatrix} = \begin{pmatrix} \boldsymbol{b}_f^{n+\frac{1}{M}}+\frac{\mathbf{M}_f}{\Delta t_f}\boldsymbol{T}_f^n+\frac{M-1}{M}\mathbf{C}_{fs}\boldsymbol{T}_s^n \\ \boldsymbol{b}_f^{n+\frac{2}{M}}+\frac{M-2}{M}\mathbf{C}_{fs}\boldsymbol{T}_s^n \\ \vdots \\ \boldsymbol{b}_f^{n+\frac{M-1}{M}}+\frac{1}{M}\mathbf{C}_{fs}\boldsymbol{T}_s^n \\ \boldsymbol{b}_f^{n+1} \\ \boldsymbol{b}_s^{n+1}+\frac{\mathbf{M}_s}{\Delta t_s}\boldsymbol{T}_s^n \end{pmatrix} \tag{14}$$

where $\mathbf{M}$ and $\mathbf{A}$ denote the mass and stiffness matrices, $\mathbf{C}$ represents the coupling matrices, and $\boldsymbol{b}$ corresponds to the source term. A similar approach is adopted in space–time FEM methods, since all "time slabs" are solved simultaneously [14].

In practice, most references in the AM literature adopt either fully staggered or semi-monolithic schemes. Algorithm 1 summarizes their respective workflows. Both algorithms start with an optional predictor step to obtain an initial estimate of the slow solution at t^{n+1}. Then, the substepping cycle begins: the fast partition is advanced in time, receiving a boundary condition from the slow partition.

In the fully staggered schemes of Cheng and Wagner [17] and Canuto and Lo Giudice [24], the fast partition is marched M times until t^{n+1}, and only then the slow partition is advanced separately, receiving a boundary condition from the last micro-step. The authors also classify Elahi et al. [18] as a fully staggered scheme; although it employs 3 subdomains, all are marched separately i.e. without any combined solve.

In the semi-monolithic schemes of Hodge [1] and Viguerie et al. [3], the fast partition is marched $M-1$ times, and then a joint monolithic solve is performed for both partitions at t^{n+1}: the corresponding linear system is given by Eq. (16) (see Algorithm 1), under the same assumptions as Eq. (14).

The advantage of the semi-monolithic approach is that the joint solve at the end of the substepping cycle ensures that transmission conditions (12) are satisfied at t^{n+1}, which is not the case for fully staggered schemes. However, the latter is computationally cheaper and more modular.

Once both partitions have been advanced to t^{n+1}, one could update the interface conditions at the micro-steps and repeat the substepping cycle, until no significant change is observed in the solutions. In practice, no scheme in the LPBF literature performs more than one substepping iteration, since reducing computational cost is the primary motivation. Hence the corresponding superscript in Algorithm 1 is omitted in the rest of this document. As it will be shown in Section 4.1.3, this truncation has a negligible impact on the accuracy of the solution.

In this work, two substepping schemes are implemented and compared: the semi-monolithic Dirichlet substepper presented in [1], and a modified version of the fully staggered Robin–Robin substepper presented in [24]. In the remainder of this document, they will be referred to as the *Dirichlet* and *Robin* substeppers, for brevity.

2.3.1. Dirichlet substepper

Fig. 5 illustrates an example step of the Dirichlet substepper. First, the predictor step (Fig. 5(b)) obtains an initial estimate of the slow

Algorithm 1 Substepping workflows

▷ *The first superscript indicates the substepping staggered iteration count.*

if `do_predictor` **then**
  $T_s^{0,n+1} \leftarrow \tilde{T}_s^{n+1}$ from predictor step over Ω at Δt_s
else
  $T_s^{0,n+1} \leftarrow T_s^n$
$i \leftarrow 0$
while not converged at Γ_{fs} and $i < i_{\max}$ **do**
  $i \leftarrow i + 1$
  Set solution and material composition to values at t^n

Fully staggered [17,18,24]

for $m = 1, \dots, M$ **do**
  Solve for $T_f^{i,n+\frac{m}{M}}$ on Ω_f at Δt_f with boundary condition from

$$T_s^{i-1,n+\frac{m}{M}} = \left(1 - \frac{m}{M}\right) T_s^n + \frac{m}{M} T_s^{i-1,n+1} \tag{15}$$

  Solve for $T_s^{i,n+1}$ on Ω_s with boundary condition from $T_f^{i,n+\frac{m}{M}}$

end while
$T_s^{n+1} \longleftarrow T_s^{i,n+1}, \qquad T_f^{n+1} \longleftarrow T_f^{i,n+1}$

Semi-monolithic [1,3]

for $m = 1, \dots, M-1$ **do** ▷ *All substeps but one*
  Solve for $T_f^{i,n+\frac{m}{M}}$ on Ω_f at Δt_f with boundary condition from Eq. (15).
  Monolithic solve for $T_s^{i,n+1}$ and $T_f^{i,n+1}$. For a linear problem with weak coupling, the system reads

$$\begin{pmatrix} \frac{\mathbf{M}_f}{\Delta t_f} + \mathbf{A}_f & \mathbf{C}_{fs} \\ \mathbf{C}_{sf} & \frac{\mathbf{M}_s}{\Delta t_s} + \mathbf{A}_s \end{pmatrix} \begin{pmatrix} \boldsymbol{T}_f^{i,n+1} \\ \boldsymbol{T}_s^{i,n+1} \end{pmatrix} = \begin{pmatrix} \boldsymbol{b}_f^{n+1} + \frac{\mathbf{M}_f}{\Delta t_f} \boldsymbol{T}_f^{n+\frac{M-1}{M}} \\ \boldsymbol{b}_s^{n+1} + \frac{\mathbf{M}_s}{\Delta t_s} \boldsymbol{T}_s^{n} \end{pmatrix} \tag{16}$$

solution at t^{n+1}, $\tilde{T}_s^{n+1}$. Then, $M-1$ substeps are performed on Ω_f: Fig. 5(c) shows the solution $T_f^{n+\frac{m}{M}}$, $1 \le m < M$ after a generic substep. The Dirichlet condition imposed on Γ_{fs} is obtained from the linear interpolation (13) using the previous slow solution T_s^n and the predicted one $\tilde{T}_s^{n+1}$:

$$T_f^{n+\frac{m}{M}} = T_s^{n+\frac{m}{M}} = \left(1 - \frac{m}{M}\right) T_s^n + \frac{m}{M} \tilde{T}_s^{n+1} \qquad \forall \mathbf{x} \in \Gamma_{fs} \tag{17}$$

From a transmission conditions standpoint, only the continuity of the solution is enforced at the substeps.

Finally, the monolithic solve (or *corrector step*) solves for both T_f^{n+1} and T_s^{n+1}. Since same mesh DD is employed, the interface DOFs are shared between the two partitions. In this work, they are assigned to the fast partition.

2.3.2. Robin substepper

An example step of the Robin substepper is also shown in Fig. 5. The predictor step is identical to the Dirichlet case. Then, M substeps are performed on Ω_f: Fig. 5(e) shows the solution at the m-th substep, $T_f^{n+\frac{m}{M}}$, $1 \le m < M$. The Robin condition imposed on Γ_{fs} reads

$$\begin{aligned} \gamma_f T_f^{n+\frac{m}{M}} + k_f \partial_n T_f^{n+\frac{m}{M}} &= \gamma_f T_s^{n+\frac{m}{M}} + k_s \partial_n T_s^{n+\frac{m}{M}} \qquad \forall \mathbf{x} \in \Gamma_{fs} \\ \gamma_s T_s^{n+1} + k_s \partial_n T_s^{n+1} &= \gamma_s T_f^{n+1} + k_f \partial_n T_f^{n+1} \qquad \forall \mathbf{x} \in \Gamma_{sf} \end{aligned} \tag{18}$$

where k_f and k_s are the thermal conductivities on the fast and slow partitions, respectively. The coefficients γ_f and γ_s are generally set to 1, see [24]; however, one can attempt to optimize this choice in order to improve the convergence of the staggered scheme, as done analytically in [17] for Dirichlet–Robin coupling. Following Chen et al. [20], in this work they are set to $\mathcal{O}(1)$ and $\mathcal{O}(k/h)$, respectively. This choice is "optimal" in the sense that it should result in mesh-independent convergence rate of the staggered scheme.

Finally, the *corrector step* solves for T_s^{n+1} on Ω_s, receiving a Robin condition from the last micro-step on Ω_f. Fig. 5(f) shows the solution T_s^{n+1} after the corrector step. Note that the discontinuities at the interface are a natural consequence of the truncated staggered scheme and the weak enforcement of solution continuity.

It is worth mentioning that the authors initially pursued a Dirichlet–Neumann variant of the substepper; however, as per Canuto and Lo Giudice [24], the authors found that it lacked both accuracy and robustness, requiring a relaxation parameter to converge (see Section 4.1.3).

2.4. Substepped advected subdomain methods

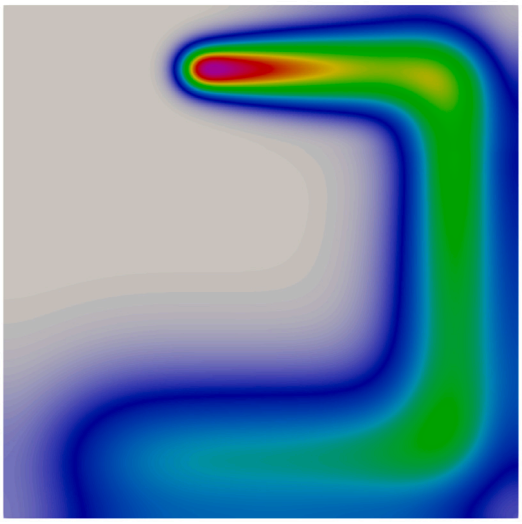
The main purpose of this work is to combine the advantages of the advected subdomain method and the substepping schemes. The moving domain Ω_m is embedded within the fast partition Ω_f, such that the substeps are carried out using the advected subdomain method. This is advantageous compared to the original advected subdomain method since it considerably reduces the size of the asymmetric system by excluding the slow partition. The fixed subproblem defined in Section 2.2 corresponds here to $\Omega_f \setminus \Omega_m$ and acts as a buffer between Ω_m and Ω_s.

As a consequence of using the steadiness workflow of the advected subdomain model, a macro-step is no longer divided into a fixed number of substeps, since Δt_f is increased adaptively. The advected subdomain will be inserted within both the Robin and Dirichlet substeppers described in Section 2.3. The resulting schemes can be regarded as three level methods, similar to Leonor and Wagner [11] and Elahi et al. [18].

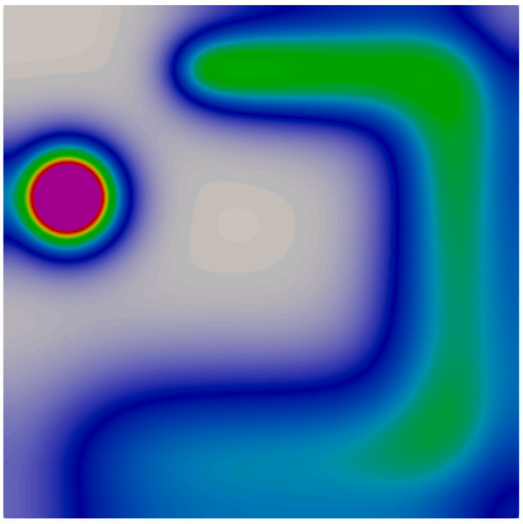
3. Implementation

The schemes presented in Section 2 were implemented on top of the `FEniCSx` software library [27]. For the solution of linear systems, the `PETSc` library was employed. Following the `FEniCSx` design philosophy, high-level functionality was written in `Python`, while performance-critical components were implemented using `C++` and exposed to `Python` via `nanobind` [28]. The code heavily relies on the `multiphenicsx` extension [29] of `FEniCSx`, which provides DOF-deactivation functionality used to implement the growth of the domain and the DD routines.

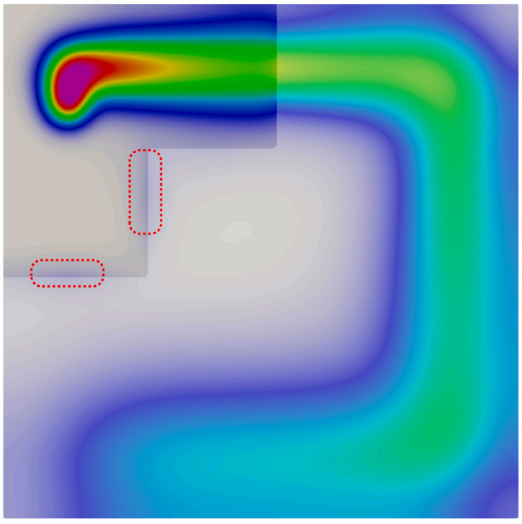
The implementation of the presented substepping strategies was relatively simple in comparison to the implementation of the advected subdomain scheme. In this work, the former are based on same-mesh DD, which is natively supported by `FEniCSx`. However, the latter relies on multi-mesh DD strategies, which required the development of custom interface detection algorithms and assembly routines for both the residual and Jacobian contributions of Eqs. (7) and (8), a task



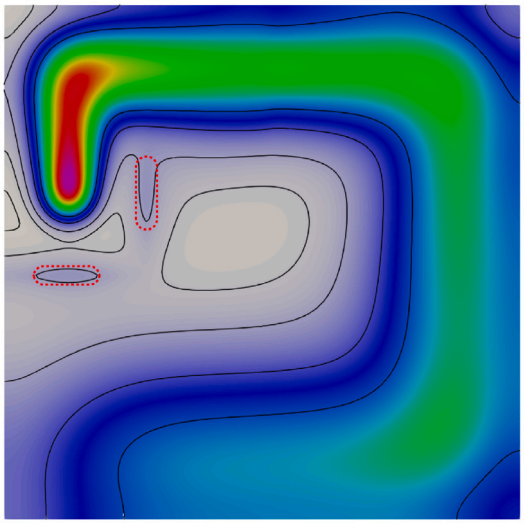
(a) Previous solution T^n.



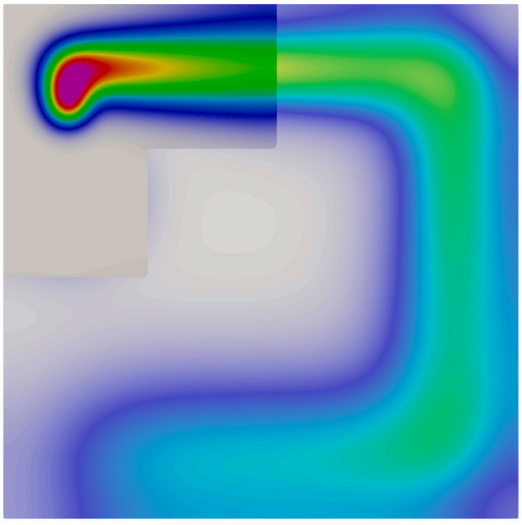
(b) $\tilde{T}^{n+1}$ after predictor step.



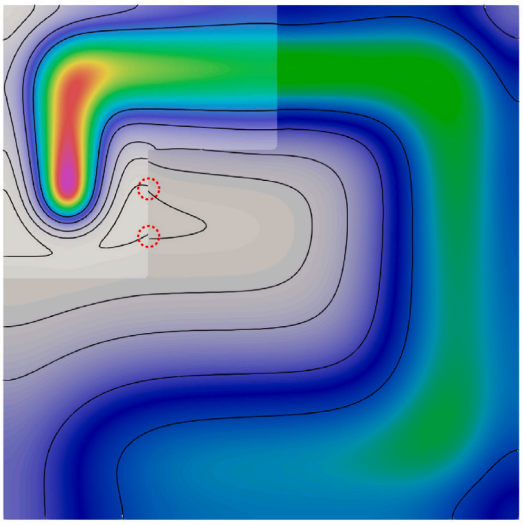
(c) Intermediate **Dirichlet** substep. The Dirichlet condition does not let the solution deviate from the predictor.



(d) **Dirichlet** corrector step. Note the leftover temperature peaks from the predictor.

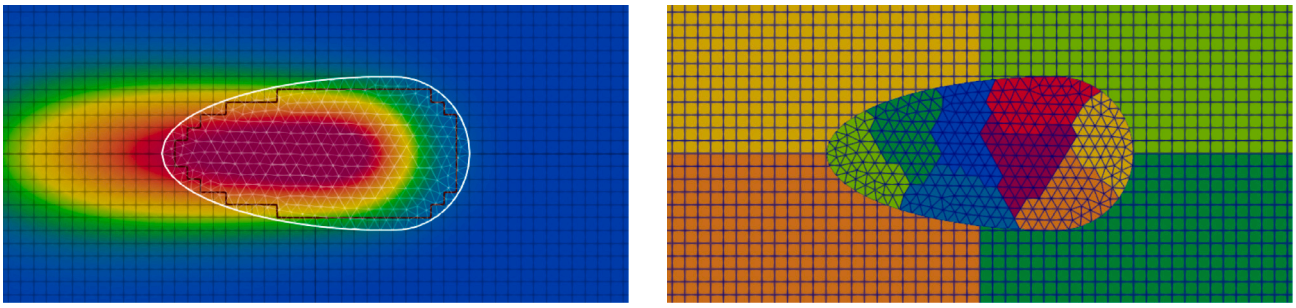


(e) Intermediate **Robin** substep. Compared to the Dirichlet substep, the solution is free to diffuse away from the predictor.



(f) **Robin** corrector step. Note the discontinuities in the solution.

Fig. 5. Examples of Dirichlet and Robin substepping cycles.



(a) Solution fields and detected interfaces Γ_m (white) and Γ_f (black). (b) Partitions of $\mathcal{T}_m$ (center) and $\mathcal{T}_f$ (outside) over 8 MPI ranks.

Fig. 6. Parallel implementation of the advected subdomain scheme based on multi-mesh DD.

further complicated by the distributed-memory setting of MPI, where coupled DOFs often reside on different processes.

Interface detection was implemented as follows: a quadrature rule is generated on all the boundary facets of $\mathcal{T}_i$, where $i = m, f$. For each Gauss point of each facet, a spatial search is performed on the cells of $\mathcal{T}_j$, $j \neq i$. The search returns ownership information in the form of the rank and local element index of the cell of $\mathcal{T}_j$ that contains the point, if any. A boundary facet of $\mathcal{T}_i$ is marked as an interface facet only if all its quadrature points are located in $\mathcal{T}_j$; otherwise it is discarded.

This procedure may produce false positives in corner cases, such as closed quadrature rules and single element protrusions of one mesh into another. Nevertheless, it is robust and efficient, as ownership information is later reused during the assembly of the coupling contributions to associate each Gauss point with the DOFs of the neighboring mesh it must couple to, and the geometry of the cell that contains it. This is necessary in a distributed memory setting, since each process stores only a subset of the global mesh and coupled DOFs of the two meshes will be owned by different processes and continuously change as $\mathcal{T}_m$ moves. Fig. 6 shows an example of a monolithic step and the corresponding meshes and partitions.

The full implementation is publicly available in the repository associated with this work [25]. A pre-built `Docker` image is also provided to facilitate reproducibility of the results presented in Section 4; see the repository's `README` for detailed instructions.

4. Numerical examples

Throughout the numerical examples, P1 and Q1 elements are used. First, the performance of the proposed substepping schemes is examined in a 2D welding case. Then, the predictive accuracy of the model is demonstrated by reproducing experimental melt pool shape results from the AMB2018-02 benchmark [30]. Finally, the methodology is applied to 3D LPBF cubes of increasing size to assess the computational speed of the various substeppers.

4.1. 2D square track

The different substeppers, their accuracy and their respective strengths and weaknesses are studied in the context of a 2D heating path problem. The problem consists of a Gaussian heat source

$$r_{(\mathbf{x},t)} = \frac{2P}{\pi R^2} \exp\left(\frac{-2\|\mathbf{x} - \mathbf{x}_{\mathbf{hs}(t)}\|_2^2}{R^2}\right)$$

moving along the square path shown in Fig. 7. Both linear and nonlinear cases are considered, with the latter accounting for phase change effects. Material parameters are listed in Table 1.

The first object of study is the temporal convergence behavior of the different substeppers. Each track is 16 R long, so the coarsest time-steps are chosen as $\Delta t_s = 16 T_{hs}$ and $\Delta t_f = 1 T_{hs}$, corresponding to one track per macro-step and Δt_f right on the bound of constraint (2). The time-steps are then successively refined in factors of 2 while maintaining the ratio $\Delta t_s / \Delta t_f = 16$, down to $\Delta t_s = 1 T_{hs}$ and $\Delta t_f = \frac{1}{16} T_{hs}$. To isolate temporal from spatial discretization errors, a fine uniform mesh with element size $h = \frac{R}{16}$ is used. The reference solution is computed with a finer discretization of $h = \frac{R}{64}$ and $\Delta t = \frac{1}{64} T_{hs}$, and does not employ substepping.

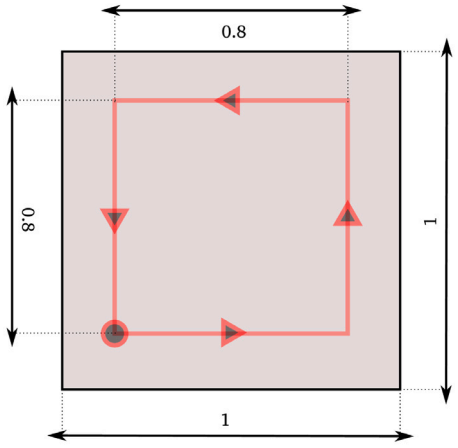


Fig. 7. Domain and heating track.

Table 1
Ti64-like material parameters for 2D square track. Phase change parameters are only used in the non-linear case.

Property	Value	Unit
Density ρ	4.3×10^{-6}	kg mm^{-3}
Specific heat c_p	700.0	J kg^{-1} °C^{-1}
Conductivity k	1.0×10^{-2}	W mm^{-1} °C^{-1}
Radius R	0.1	mm
Power P	500.0	W
Speed $\|\|\dot{\mathbf{x}}_{\mathbf{hs}}\|\|$	1000.0	mm s^{-1}
Convection coeff. h_{conv}	10	W mm^{-1} °C^{-1}
Solidus temperature T_s	1600.0	°C
Liquidus temperature T_l	1650.0	°C
Latent heat L	350.0	kJ kg^{-1}
Smoothing constant S	0.5	

Fig. 8 reports the L2 error of the solution over the entire domain at the end of the first and fourth tracks in the top and bottom rows, respectively. The substeppers considered are the Robin and Dirichlet substeppers with and without advected subdomains. The solution without substepping and $\Delta t = \Delta t_f$ is included for comparison.

All the substeppers achieve a convergence rate slightly below 1, while the unsubstepped solution achieves a rate above 1. In all cases,

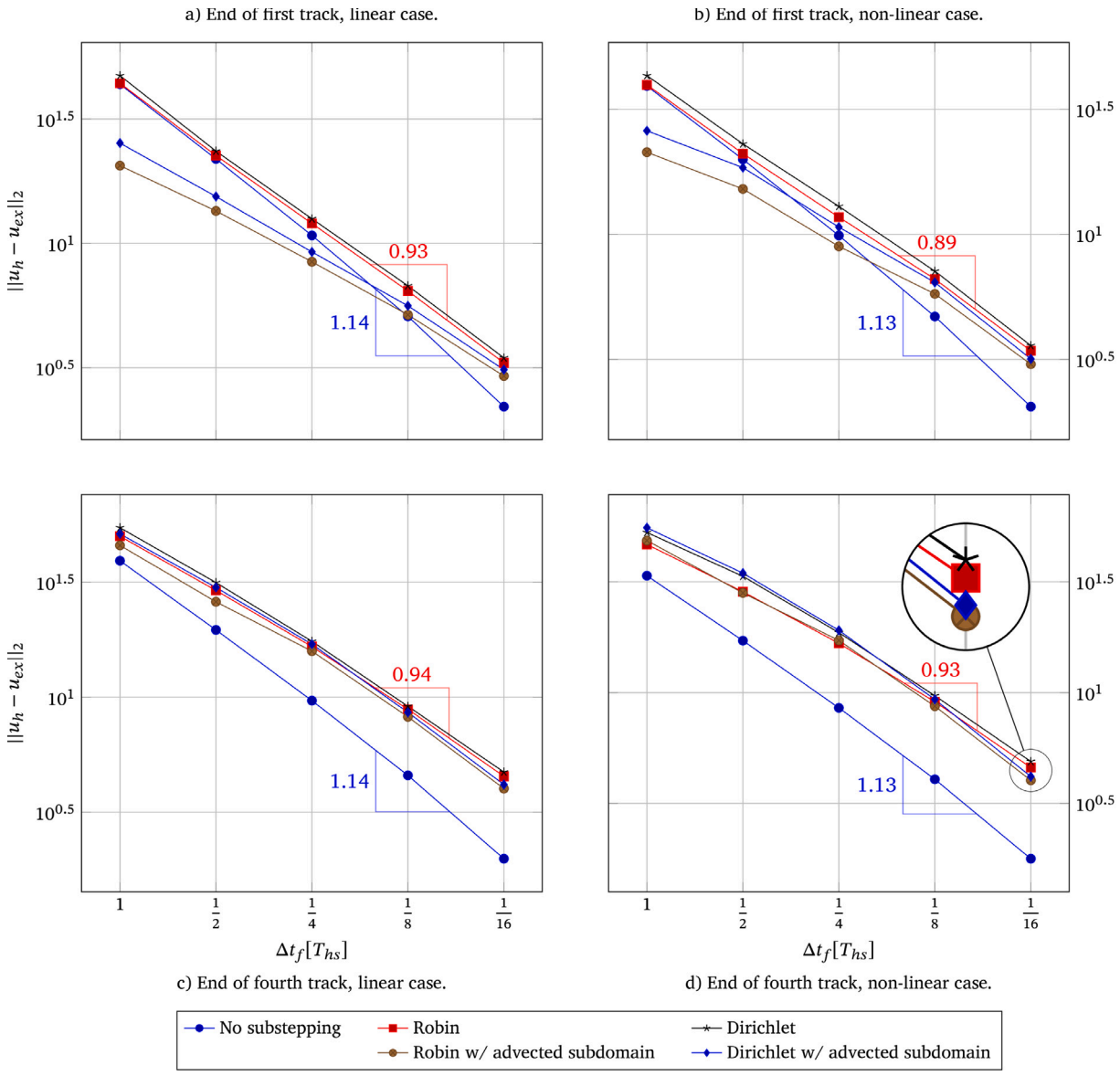


Fig. 8. Convergence of the L^2 error at the end of the first (top) and fourth tracks (bottom) for both the linear (left) and non-linear (right) cases. The horizontal axis shows Δt_f, with $\Delta t_s/\Delta t_f = 16$ kept fixed. Results are shown for all substepping strategies (Robin and Dirichlet, with and without advected subdomain), together with the unsubstepped solution for reference.

the Robin models perform slightly better than their Dirichlet counterparts in terms of error at a given time-step.

Big differences in performance are only observed at the end of the first track, where the models with advected subdomains vastly outperform the other models, especially for coarse discretizations. This is due to their higher accuracy in the HAZ.

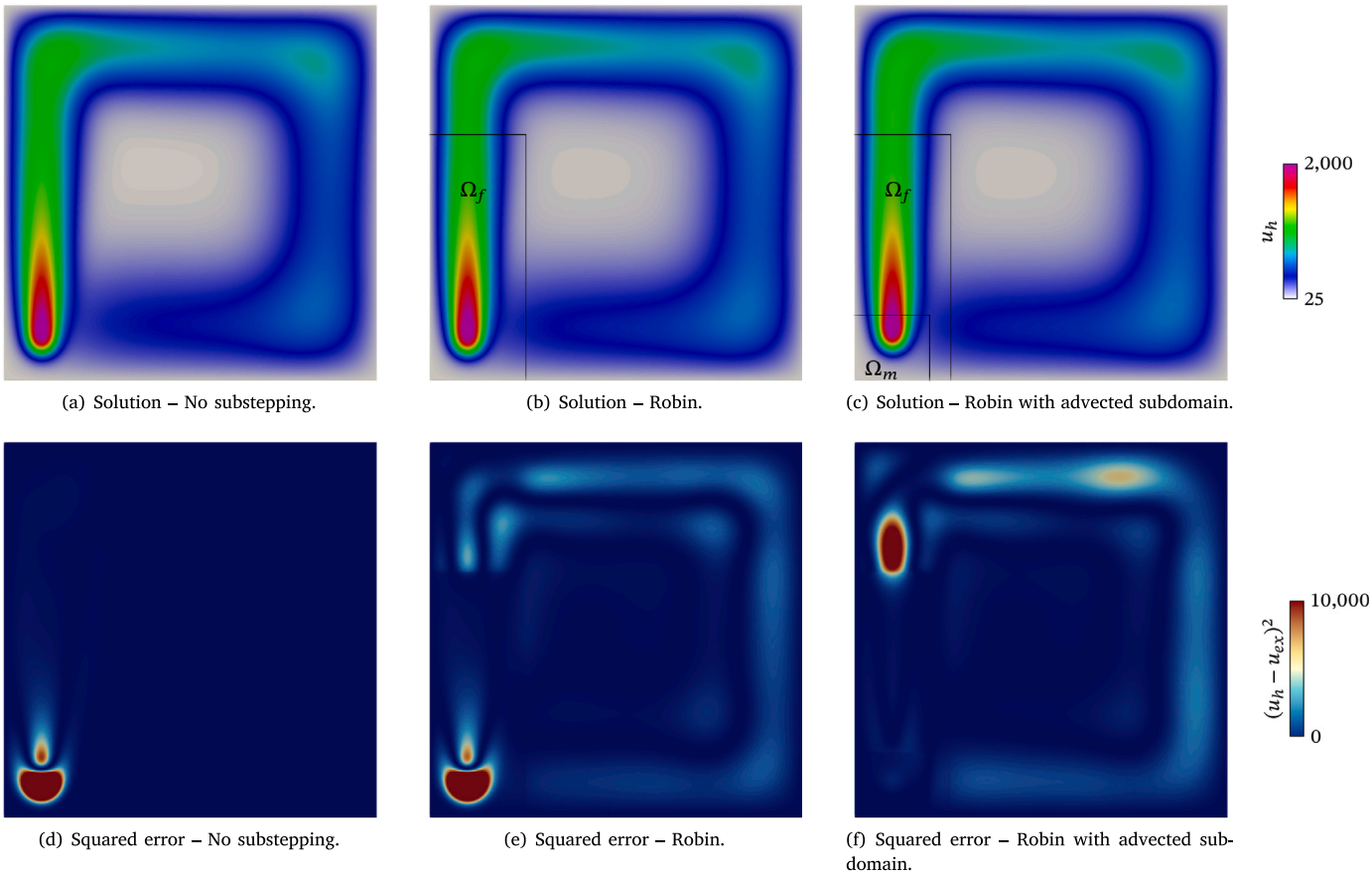
Nevertheless, their performance degrades over time due to their poor handling of sharp turns, as illustrated in Fig. 9 which compares solution and error contours for the Robin substepper with and without an advected subdomain. In particular, Fig. 9(f) shows a larger error near the rear of Ω_f compared with Fig. 9(e).

Leaving the advected subdomain on at all times would shear the solution and introduce large errors [19]. Hence, the advected subdomain is turned off at each turn and re-incorporated once the solution has stabilized, typically after a few substeps. However, these switches back and forth lead to an accuracy loss right after each turn, which progressively accumulates over time. By the fourth track, the advantage of the advected subdomain has nearly vanished, as shown in Figs. 8(c) and 8(d). Although the models with advected subdomains initially outperform the other models after the first track, their performance becomes comparable by the end of the fourth track.

Nonetheless, the bottom row of Fig. 9 clearly demonstrates the benefit of the advected subdomain: the error in the HAZ is much lower than in the unsubstepped solution. This error manifests as a slight forward shift of the melt pool in the latter compared to the former, an effect that could be compensated through calibration of a double-ellipsoidal heat source model [31].

4.1.1. Comparing predictors

Comparing Figs. 5(c) and 5(e), one remarks that the Dirichlet substepper is much more sensitive to the predictor than the Robin substepper. In particular, observe on Fig. 5(d) the spurious temperature peaks at the interface between the fast and slow regions. These arise because the heat source reaches Γ_{fs} during the predictor step in Fig. 5(b); the Dirichlet condition preserves this excess heat in subsequent substeps, whereas in the Robin substepper it is allowed to diffuse away.



(a) Solution – No substepping. (b) Solution – Robin. (c) Solution – Robin with advected subdomain.

(d) Squared error – No substepping. (e) Squared error – Robin. (f) Squared error – Robin with advected subdomain.

Fig. 9. Solutions (top row) and pointwise squared errors (bottom row) in the linear case for the Robin substepper with and without an advected subdomain, with $\Delta t_s = 8T_{hs}$ and $\Delta t_f = 0.5T_{hs}$. The unsubstepped solution with $\Delta t = 0.5T_{hs}$ is shown for reference. The substeppers introduce some error throughout the domain. The advected subdomain reduces the error in the HAZ but increases the error near the track turns.

Since the diameter of the heat spot produced by the predictor is proportional to Δt_s, one can either decrease Δt_s or enlarge Ω_f to mitigate the issue, although both approaches can increase the computational cost.

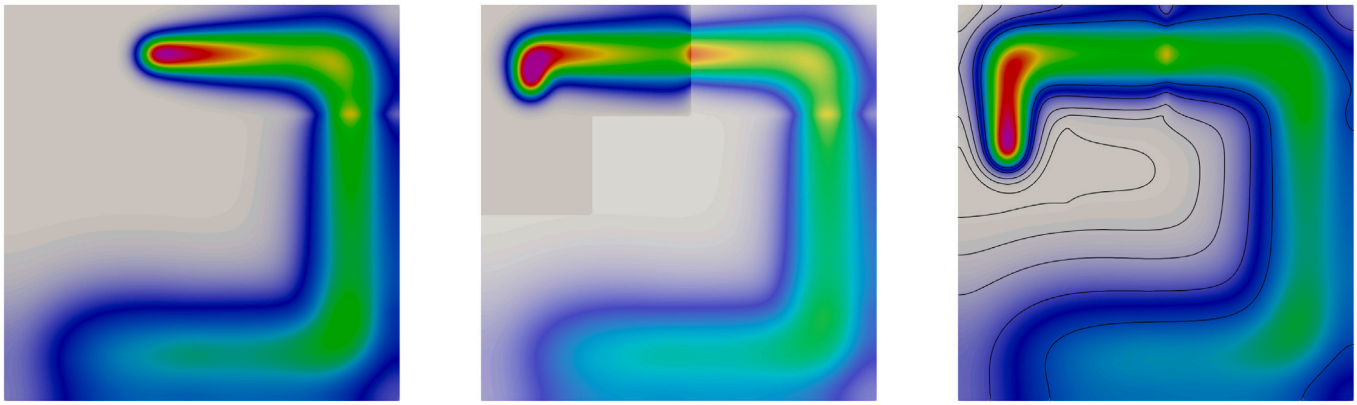
Since the macro-step size Δt_s is typically much larger than $1T_{hs}$, the predictor step cannot possibly resolve the heat source, making its heating profile unreliable. Nevertheless, a predictor step remains essential as indicated by Fig. 10, which illustrates a Dirichlet substepping cycle without a predictor step. The resulting solution is unsatisfactory, as the diffusion at the intersection of Γ_{fs} and the heating track is severely underestimated. Skipping the predictor step is equivalent to using the previous solution as the predictor, which amounts to evaluating the interface terms at t^n, effectively introducing an explicit component in their time integration [8]. Furthermore, in the case of the Dirichlet substepper, the substeps remain constrained to the solution at t^n along the interface, leading to the observed stagnation.

This analysis suggests that the predictor should focus on capturing diffusive effects, whose characteristic time scale should be on the order of Δt_s. To explore this, two alternative predictor strategies are considered: (1) **turning off the source term** (Fig. 11) and (2) using a **lumped heat source** profile (Fig. 12). The former completely foregoes heating, while the latter attempts to approximate heating effects while still employing a large time-step, as is common in AM simulations [8, 23,32]. Standard collision detection algorithms are used to identify the elements affected by the lumped heat source, with a collision box extending $1R$ in the directions transverse to the heat source motion, and delimited by the positions of the heat source at t^n and t^{n+1} along the track direction.

Fig. 13 compares the three predictor strategies: standard, heat source (HS) turned off and lumped HS. The left panel shows the convergence rates of the three predictors in the linear case for both the Robin and Dirichlet substeppers. The results are reported without the advected subdomain at the end of the fourth track. As expected, the predictor has negligible effect on the Robin substepper. In contrast, the Dirichlet substepper is significantly more sensitive to the predictor choice, especially for coarser time steps Δt_s. This sensitivity decreases as Δt_s is decreased, since the predicted heat output is more localized and no longer reaches Γ_{fs}.

The lumped HS predictor improves the results compared with the standard predictor, but turning off the HS completely is clearly the better option. The right panel of Fig. 13 compares solution and error contours for the standard and no-HS predictors. Similarly to Figs. 5(d), 13(c) shows residual heating at Γ_{fs} due to the predictor, which translates to elevated error in the same region. These features are absent from Figs. 13(b) and 13(d), which achieve a lower error altogether.

An advantage of the no-HS and lumped HS predictors over the standard predictor is that they yield lower temperatures, which fare better with non-linear iterations: as noted in Puso and Hodge [16], the high temperatures imposed by the standard predictor can cause convergence issues in the non-linear iterations.

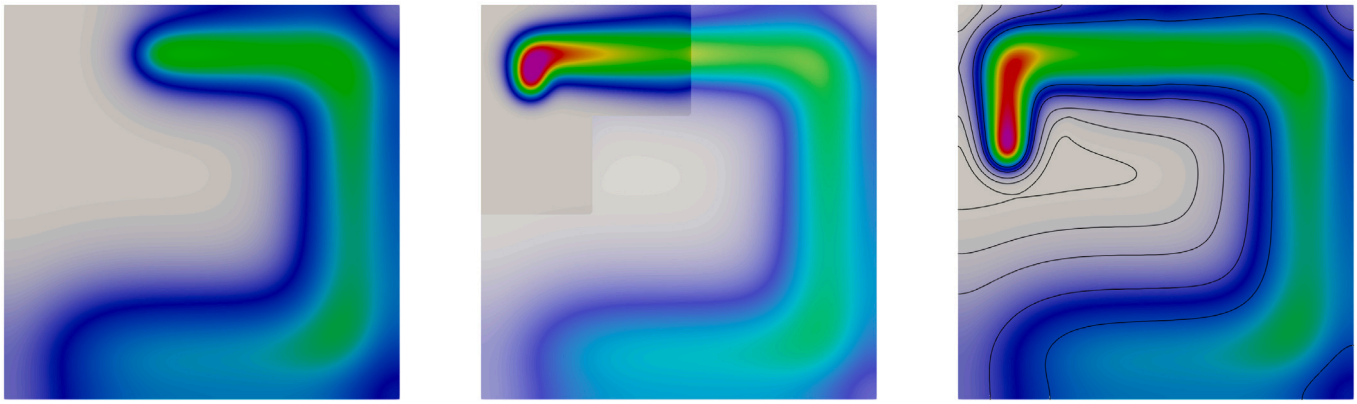


(a) Previous solution T^n, which also serves as the predictor $\tilde{T}^{n+1}$.

(b) Intermediate substep.

(c) Corrector step.

Fig. 10. Example of Dirichlet substepper step without a predictor step.

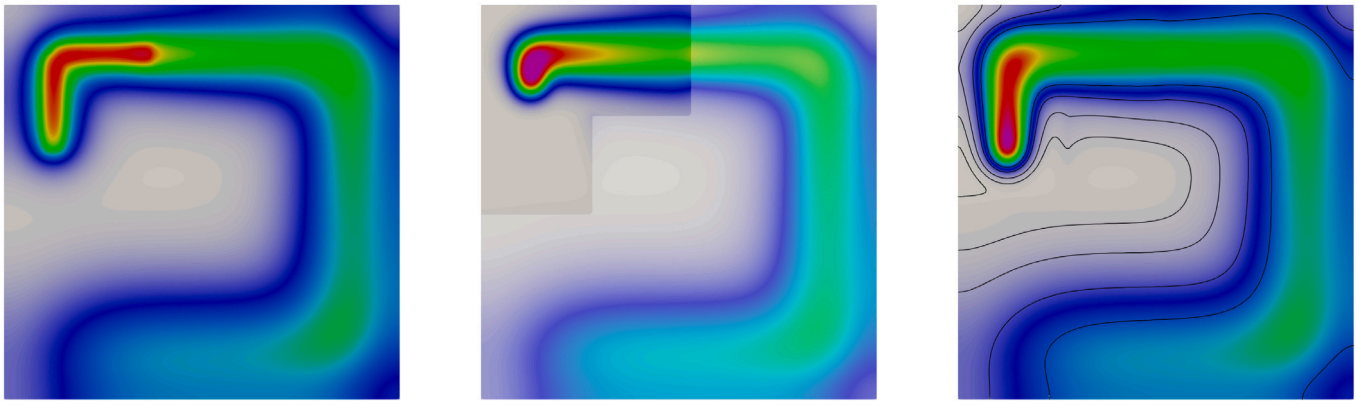


(a) Predictor step without heat input

(b) Micro-step.

(c) Corrector step.

Fig. 11. Example of Dirichlet substepper step using a predictor step excluding the source term.



(a) Predictor step with lumped heat source.

(b) Micro-step.

(c) Corrector step.

Fig. 12. Example of Dirichlet substepper step using a lumped heat source predictor.

4.1.2. On the effect of Δt_s

It is important to understand the cost-accuracy trade-off associated with increasing Δt_s. In the authors' experience, it often leads to a jump at the fast-slow interface. To accentuate this effect, the conductivity is halved in the following examples to reduce the diffusion of this jump. The advected subdomain is omitted, as it is not relevant to the discussion.

Fig. 14 shows the solution profiles along the first heating track for $\Delta t_s = 4, 8, 16 T_{hs}$, for both the Dirichlet and Robin substeppers. A fine Δt_f of $\frac{1}{8} T_{hs}$ is kept constant to isolate the effect of Δt_s. For both the linear and non-linear cases, increasing Δt_s leads to progressively larger

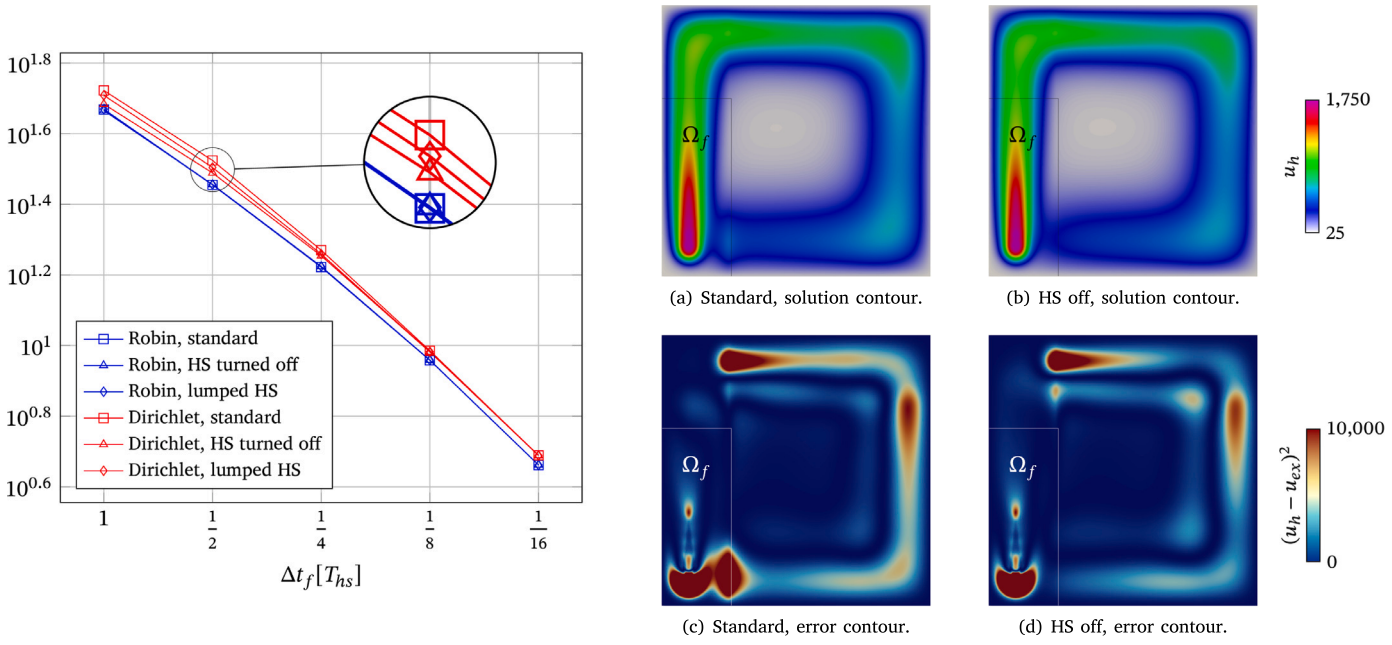


Fig. 13. Comparison of predictor strategies.
Left, convergence rates in the linear case for the standard, the heat source (HS) turned off and the lumped HS predictors.
Right, solutions and errors of the Dirichlet substepper at the end of the fourth track in the non-linear case, using both the standard and the HS turned off predictors, with $\Delta t_s = 16T_{hs}$ and $\Delta t_f = 1T_{hs}$.

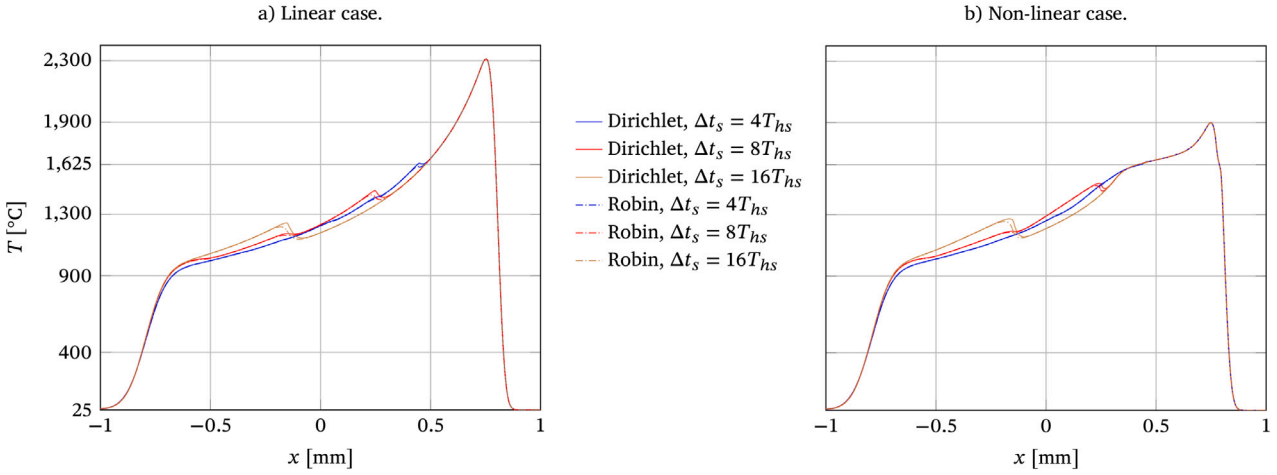


Fig. 14. Solution profiles along the first heating track for increasing values of Δt_s and constant $\Delta t_f = \frac{1}{8}T_{hs}$, for both the Dirichlet and Robin substeppers without the advected subdomain. The no-HS predictor is employed.

jumps at the fast-slow interface, with the slow side lagging behind the fast side and overpredicting the temperature in the thermal tail.

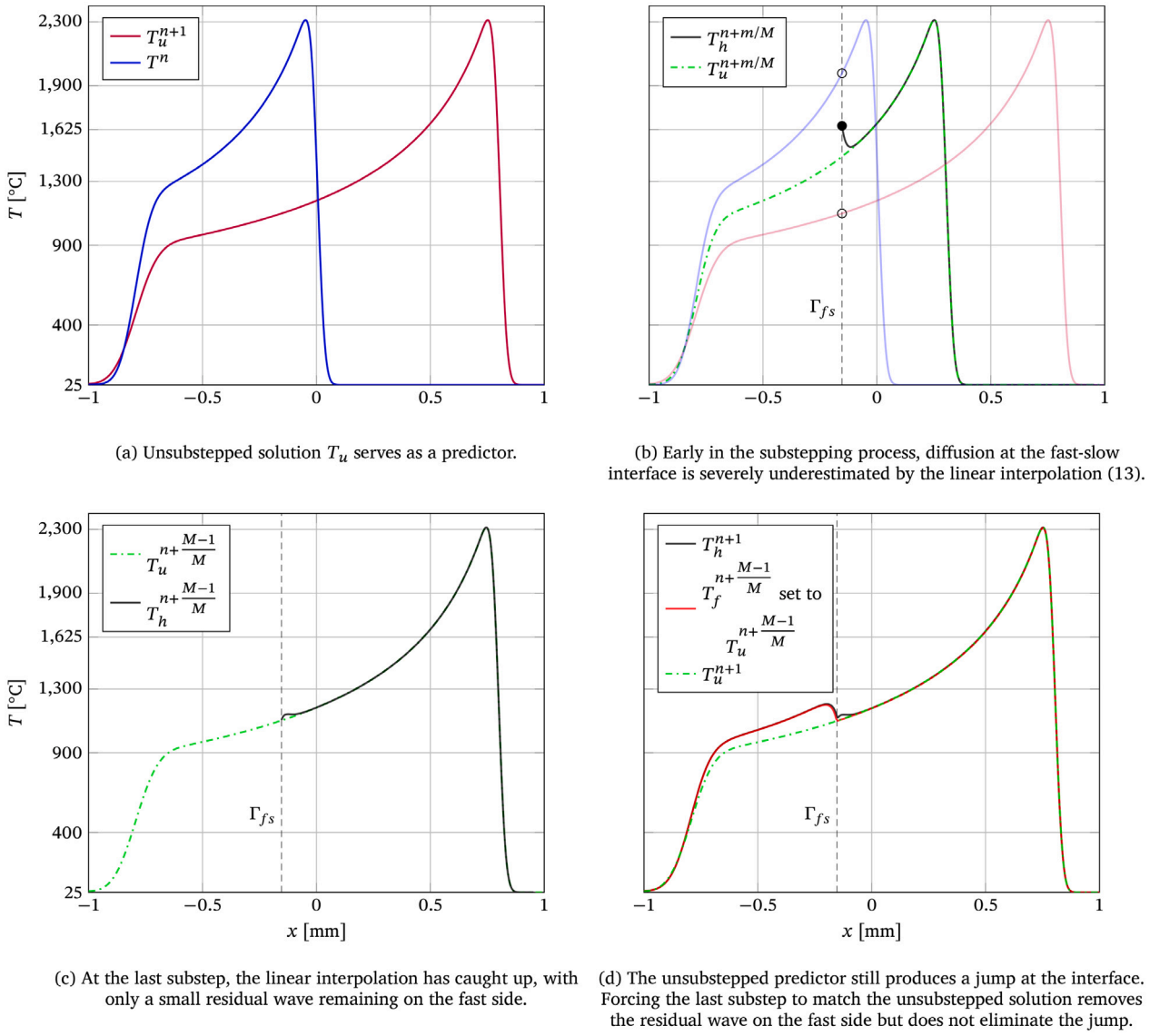
To further isolate the effect of Δt_s, the unsubstepped solution is fed as a predictor to the Dirichlet substepper in Fig. 15. Fig. 15(b) shows that the linear interpolation of the Dirichlet condition (13) severely underestimates the diffusion at the interface. This occurs because diffusion is strongest when the region is hotter than its surroundings, and gradually slows down as thermal equilibrium is approached, a deceleration that the linear interpolation cannot capture. By the last substep, the linear interpolation (13) catches up, and only a small residual wave remains on the fast side, as shown in Fig. 15(c).

Still, the jump at the interface persists despite the good predictor, as shown in Fig. 15(d). Forcing the last substep to the unsubstepped solution does not remove the jump, although it does remove the residual wave on the fast side caused by the linear interpolation. By exclusion, the jump must be due to the poor time integration on the slow side, which underestimates the diffusion in Ω_s.

Note that the issue is encountered at around $\Delta t_s = 5T_{hs}$. As discussed in Section 1, no reference in the literature goes over $\Delta t_s = 10T_{hs}$, with most studies operating near $\Delta t_s = 5T_{hs}$. Since these works, like the present one, employ first-order time integration on the slow side, the authors conclude that the current bottleneck on employing larger Δt_s is the time integration on the slow side. Employing higher-order time integration schemes may offer a way forward: although an increase in computational cost is expected, each macro-step advances the solution over a large time interval, maximizing the impact of improved accuracy.

4.1.3. *On the effect of truncated staggered iterations*

As explained in Section 2.3, the presented substepping schemes can be seen as staggered DD schemes truncated at the first iteration.



(a) Unsubstepped solution T_u serves as a predictor.

(b) Early in the substepping process, diffusion at the fast-slow interface is severely underestimated by the linear interpolation (13).

(c) At the last substep, the linear interpolation has caught up, with only a small residual wave remaining on the fast side.

(d) The unsubstepped predictor still produces a jump at the interface. Forcing the last substep to match the unsubstepped solution removes the residual wave on the fast side but does not eliminate the jump.

Fig. 15. Step-by-step illustration of a linear problem where the Dirichlet substepper uses the unsubstepped solution as a predictor, with $\Delta t_s = 16T_{hs}$ and $\Delta t_f = \frac{1}{8}T_{hs}$.

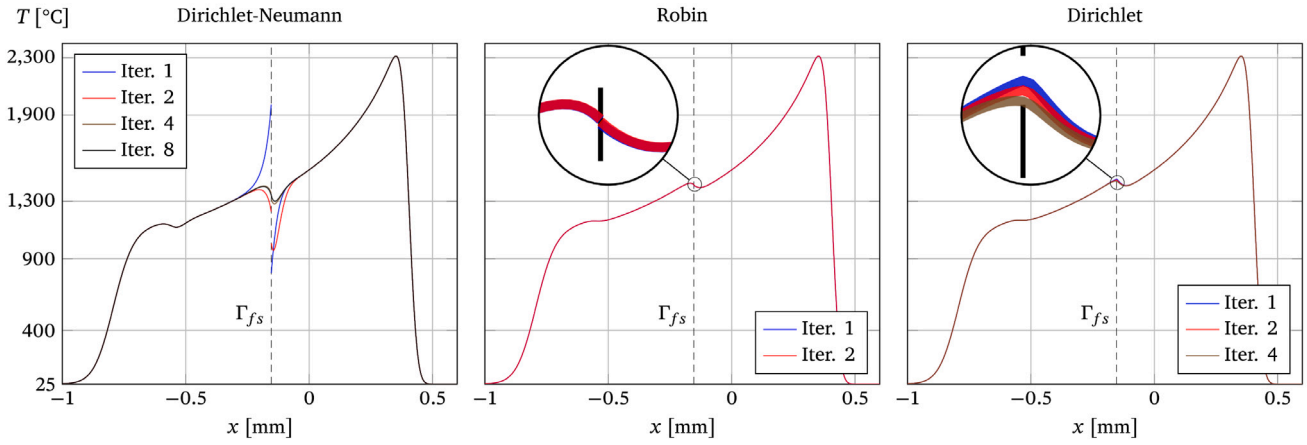


Fig. 16. Solution profiles along the heating track after multiple staggered iterations for a linear problem with $\Delta t_s = 8T_{hs}$ and $\Delta t_f = 0.125T_{hs}$. Substeppers from left to right: Dirichlet–Neumann with Aitken relaxation, Robin, and Dirichlet.

It is therefore natural to investigate whether this early truncation significantly impacts solution accuracy.

Fig. 16 shows solution profiles along the heating track for multiple substeppers. The left-most plot corresponds to a fully staggered

Table 2
Material properties for IN625 used for FEM.
Source: From [14,30].

Properties	Value	Unit
Solidus temperature T_s	1290	°C
Liquidus temperature T_l	1350	°C
Latent heat L	280.0	kJ kg^{-1}
Density ρ	8440.0	kg m^{-3}
Specific heat c_p	{(0.0, 405), (1320, 723.6), (2000, 723.6)}	(°C, J kg^{-1} °C^{-1})
Conductivity k	{(0.0, 9.5), (1320, 28.85), (2000, 28.85)}	(°C, W m^{-1} °C^{-1})

Dirichlet–Neumann scheme. Without relaxation the scheme diverges, so Aitken relaxation [33] is applied. Even with relaxation, convergence is slow, requiring more than 10 iterations.

The central plot shows the Robin substepper, which practically shows no change between the first and second iterations. The coefficient choice proposed by Chen et al. [20] appears to perform well in this case. The magnified interface region reveals only a very small jump in the solution across Γ_{fs}, which is consistent with the weakly imposed Robin condition. These results corroborate the findings of Canuto and Lo Giudice [24] and Cheng and Wagner [17], which report that the Robin substepper vastly outperforms the Dirichlet–Neumann one.

The right-most plot shows the Dirichlet substepper, which converges within a few iterations. The solution initially exhibits a peak at the interface which progressively smooths out as the iterations proceed. Nevertheless, the difference between the first and last iterations is relatively small.

Overall, the results suggest that truncating the staggered iterations at the first iteration does not significantly impact accuracy, provided that a suitable substepper is chosen.

4.2. AMB2018-02

As part of the 2018 Additive Manufacturing Benchmark test series (AM-Bench 2018), Lane et al. [30] conducted a series of single-track experiments on bare IN625 substrates, varying the laser power and the scanning speed and recording the resulting melt pool dimensions. In this section, the model presented in Section 2.1 is validated against 3 of those experiments, namely, the ones conducted on the AMMT machine and thermally imaged every 20 µs. These correspond to the last three rows of Table 3 of Lane et al. [30] and are denoted tracks A2 ($V = 0.4$ m s^{-1}, $P = 137.9$ W), B3 ($V = 0.8$ m s^{-1}, $P = 179.2$ W) and C1 ($V = 1.2$ m s^{-1}, $P = 179.2$ W).

Kopp et al. [14] performed a similar experimental validation, focusing on track B3. The material parameters reported in [14] are also adopted in this work (see Table 2), along with the same heat-source model:

$$r(\boldsymbol{\xi}) = \frac{\nu P}{2\pi R^2} \exp\left(-\frac{\xi_x^2 + \xi_y^2}{2R^2}\right) \frac{2}{\sqrt{2\pi} R_z} \exp\left(-\frac{\xi_z^2}{2R_z^2}\right) \tag{19}$$

which corresponds to a two-dimensional Gaussian of radius R extended along the depth direction by another Gaussian of radius R_z that governs the exponential decay of the heat source with depth.

Kopp observes that "the melt pool reaches steady-state after at most 2 mm". The authors can corroborate this observation and therefore reduce the problem size to track lengths of 2 mm over substrates of dimensions 4.25×2.05×1.755 mm^3. Furthermore, the effects of radiation and convection are neglected since they operate on much larger time scales than those of heating and diffusion (s versus µs) and therefore have a negligible influence on the melt pool shape during the laser scan.

Results are reported using a fine element size of $h = \frac{1}{8}R$ in the vicinity of the melt pool, and coarser elements elsewhere (Fig. 18(a)). Coarser meshes with $h = \frac{1}{4}R$ are considered for calibration: the melt pool width and depth are practically unaffected, and the melt pool length differs by less than 2%. The slow time-step is fixed to $\Delta t_s = 6T_{hs}$, while Δt_f is set to $2T_{hs}$ and $0.25T_{hs}$ in the cases with and without an advected subdomain, respectively. The advected subdomain is well suited to the task, since the object of study is the steady state melt pool shape. From the conclusions of the previous section, it should provide more accurate results in the HAZ.

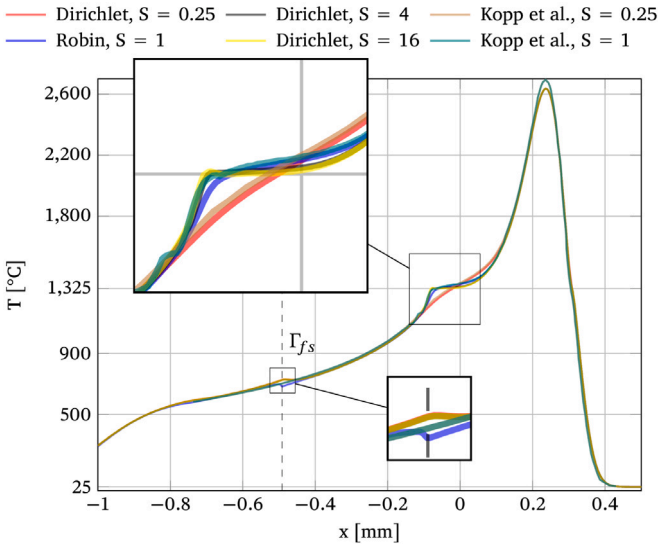


Fig. 17. Temperature along the heating track for varying S using both the Robin and Dirichlet substeppers without an advected subdomain. The dashed line indicates the location of the fast-slow interface Γ_{fs}.
Source: Results from [14] are included for reference.

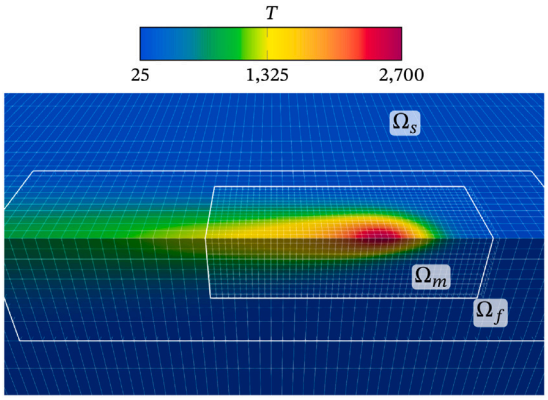
First, the results of Kopp et al. [14] are reproduced for experiment B3. Since the model of Kopp et al. employs the same phase change regularization function as in the present work, the same parameters must be calibrated: ν, R_z and S. Kopp et al. report $\nu = 0.32$ and $R_z = 0.28R$ based on calibration against the melt pool width and depth, while S is chosen to fit the melt pool length.

Fig. 17 shows the temperature along the heating track on the top surface of the substrate, using $\nu = 0.32$ and $R_z = 0.28R$ and $S \in \{0.25, 1, 4, 16\}$. Both the Dirichlet and Robin substeppers without an advected subdomain are considered, and the results are compared with those of Kopp et al. for $S = 0.25$ and $S = 1$.

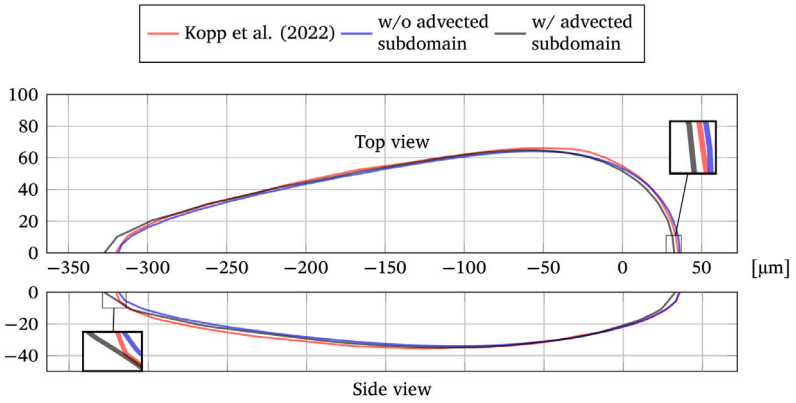
Increasing S produces a flatter temperature profile around the melting point, as expected of a more isothermal melting. This, in turn, leads to a longer melt pool (Table 3), while the width and depth remain nearly unchanged. Away from the heat source, the results are otherwise very similar, except for oscillations at the melting front observed for $S = 1$ in Kopp et al.. The present model remains stable up to $S = 16$, approaching near isothermal melting without oscillations. This difference is attributed to the source-based latent heat treatment adopted in this work, as opposed to the apparent heat capacity method employed by Kopp et al.. A detailed comparison is provided in Appendix B.

Results obtained with the Dirichlet and Robin substeppers are very similar, except at the fast-slow interface, where they exhibit slightly different oscillations in the same fashion as Fig. 14(b). These are negligible, particularly since they occur far away from the melt pool. As the results in the HAZ are effectively identical, the Robin substepper is adopted for the remainder of this section

Kopp et al. reports $S = 0.25$ as the best fit for the melt pool length, a relatively low value that is consistent with Figure 5 of Lane et al.. Both the models with and without an advected subdomain obtain similar results for $S = 0.25$, although the advection produces a slightly longer melt pool length, as shown in Table 3 and Fig. 18(b). In addition, the



(a) Intermediate substep of Robin substepper with an advected subdomain.



(b) Top and side views of simulated melt pool contours, with and without an advected subdomain, compared against results from Kopp et al. (2022).

Fig. 18. Solution and comparison with Kopp et al. [14] of melt pool contours for experiment B3.
Left: Temperature field at an intermediate substep of the Robin substepper with an advected subdomain. Right: Comparison of simulated melt pool contours.
Source: Data from Kopp et al. obtained from https://gitlab.com/hpfem/publications/2021_spacetime_am/-/releases/Submission.

Table 3
Simulated and experimental melt pool dimensions (length L, width W, depth D) for tracks B3, C1 and A2. Calibrated parameters obtained for B3 fail to generalize to C1 and A2, and new absorptivities ν_{C1} and ν_{A2} are calibrated for these cases.

Case	Dataset	P [W]	V [mm s^{-1}]	Calibrated parameters	Advected subdomain?	S	L [μm]	W [μm]	D [μm]
B3	Simulation	179.2	800.0	S, ν, R_z	✓	0.25	360.5	129.7	34.9
				S, ν, R_z	✗	0.25	354.7	128.3	34.0
				ν, R_z	✗	1	388.0	128.3	34.0
				ν, R_z	✗	4	396.5	128.3	33.9
				ν, R_z	✗	16	398.5	128.3	33.9
	Kopp et al.			S, ν, R_z	✗	0.25	356	129	34.8
	Experiment			–	–	–	359	132	36
C1	Simulations	179.2	1200.0	–	✗	0.25	298.8	106.8	24.6
				$\nu_{C1} = 0.366$	✓	0.25	362.6	117.9	28.5
	Experiment			–	–	0.25	370	113	29
A2	Simulations	137.9	400.0	–	✗	0.25	315.3	145.4	44.1
				$\nu_{A2} = 0.308$	✓	0.25	305.2	143.5	43.5
	Experiment			–	–	0.25	300	152	43

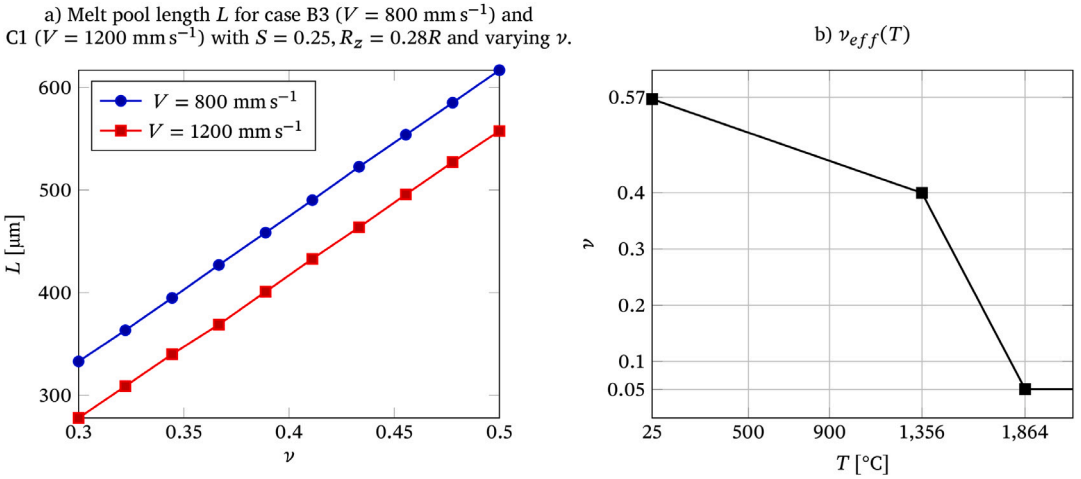


Fig. 19. Left: Melt pool length as a function of absorptivity ν for cases B3 and C1. The simulations consistently predict a longer melt pool for B3 than for C1, in contrast with the experiments, which show the opposite trend.
Right: Calibrated piecewise linear absorptivity $\nu_{eff}(T)$ found by the joint calibration on B3 and C1.

melt pool is shifted slightly backward relative to the heat source motion when advection is included.

Using an advected subdomain with $S = 0.25, \nu = 0.32, R_z = 0.28R$ achieves slightly better results in predicting L, although the improvement is small, with all predictions lying within 1.2% of the experimental results. However, this parameter set fails to generalize to the other experiments, as shown by Table 3. They lead to length prediction errors of 5% and 19.2% for tracks A2 and C1, respectively. These errors can be reduced to 0.5% and 2% by calibrating case-specific

Table 4
Results for experiments B3, C1 and A2 with piecewise linear absorptivity $\nu_{eff}(T)$. Parameters found by joint calibration on B3 and C1 manage to generalize to A2.

Case	Dataset	P [W]	V [mm s^{-1}]	Calibrated parameters	Advected subdomain?	L [µm]	W [µm]	D [µm]
B3	Simulation	179.2	800.0	$S, \nu_{eff}(T), R_z$	✓	362.5	136.6	35.6
	Experiment			–	–	359	132	36
C1	Simulation	179.2	1200.0	$S, \nu_{eff}(T), R_z$	✓	366.2	118.5	29.8
	Experiment			–	–	370	113	29
A2	Simulation	137.9	400.0	–	✓	289.3	148.9	41.4
	Experiment			–	–	300	152	43

absorptivities $\nu_{A2} = 0.308$ and $\nu_{C1} = 0.366$, while keeping S and R_z fixed.

More importantly, the experiments measure a *longer* melt pool in C1 than in B3. They both apply the same laser power, with C1 going 1.5 times faster than B3. In contrast, the presented model predicts a *shorter* melt pool in C1 than in B3. Fig. 19(a) plots melt pool lengths for absorptivities $\nu \in [0.3, 0.5]$: for a given absorptivity, the melt pool length is always shorter in C1 than in B3.

Since a longer melt pool is observed experimentally in C1, the model is most likely overestimating temperatures in B3, indicating that an energy sink is missing in the model. As noted previously, convection and radiation cannot be the cause, as they act on much longer time scales.

To compensate for the missing energy sink, a temperature-dependent absorptivity is introduced:

$$\nu_{eff}(T) = \begin{cases} \nu_0 + \frac{\nu_1 - \nu_0}{T_1 - T_0}(T - T_0) & T < T_1 \\ \nu_1 + \frac{\nu_2 - \nu_1}{T_2 - T_1}(T - T_1) & T_1 \le T < T_2 \\ \nu_2 & T \ge T_2 \end{cases}$$

This corresponds to a two-segment piecewise linear function with a constant value ν_2 past a threshold temperature T_2.

Recognizing that multiple parameter sets can produce similar melt pool shapes, a joint calibration is performed on two experiments to reduce the risk of overfitting to a single experiment. Tracks B3 and C1 are chosen, since they exhibit the qualitative behavior missing from the model.

The calibration proceeds in two steps. First, a least-squares optimization loop is carried out to determine $\nu(T)$ and S, with additional weight assigned to the melt-pool length. Then, a second optimization loop is used to determine R_z, emphasizing the width and depth. The resulting parameters are $S = 0.31$, $R_z = 0.42R$ and the piecewise linear absorptivity shown in Fig. 19(b). Interestingly, the calibration produces an absorptivity dip past the melting point: a physically unrealistic feature which, in this case, compensates for the missing energy sink.

Results obtained with the newly calibrated parameters are shown in Table 4. While the results are less accurate for B3, the errors remain well within 5% of the experimental measurements. Similar accuracy is achieved for C1, and, critically, the model now predicts a longer melt pool length in C1 than in B3. The predictions for A2 also remain within 5%.

Overall, the joint calibration is deemed successful: the model reproduces both the qualitative dependence of the melt pool length on the scanning speed and the quantitative melt pool dimensions of the independent test case A2, which was not included in the calibration.

4.3. 3D LPBF

To assess performance in a controlled yet representative setting, the LPBF print of a cube geometry is considered. Similarly to Puso and Hodge [16], the cube size is varied from 1.0 mm to 3.0 mm to study scaling effects and analyze the impact of domain size on the efficiency of the substepping schemes.

The material considered is Ti-6Al-4V with thermally dependent density and conductivity. The adopted values are reported in Table 5 and Fig. 21. Although specific values vary across studies, and there is little data available for the powder phase, the chosen parameters are representative of those commonly employed in the literature.

Table 5
Ti-6Al-4V material parameters for 3D LPBF cube simulations.

Property	Value	Unit
Solidus temperature T_s	1600	°C
Liquidus temperature T_l	1650	°C
Latent heat L	350.0	kJ kg^{-1}
Environment temperature T_{env}	25	°C
Emissivity ϵ	0.35	
Convection coeff. h_{conv}	10	W m^{-2} °C^{-1}
Specific heat c_p	560	J kg^{-1} °C^{-1}
Density ρ	Fig. 21(a)	
Conductivity k	Fig. 21(b)	

The heat input is modeled using a Gaussian profile

$$r(\mathbf{x},t) = \frac{6\sqrt{3}P}{\pi^{3/2}R^3} \exp\left(\frac{-3\|\mathbf{x} - \mathbf{x}_{\mathbf{hs}(t)}\|_2^2}{R^2}\right) \qquad \mathbf{x}_{\mathbf{hs}} = \mathbf{x}_{\mathbf{0}} + \mathbf{V}_{(t)}t$$

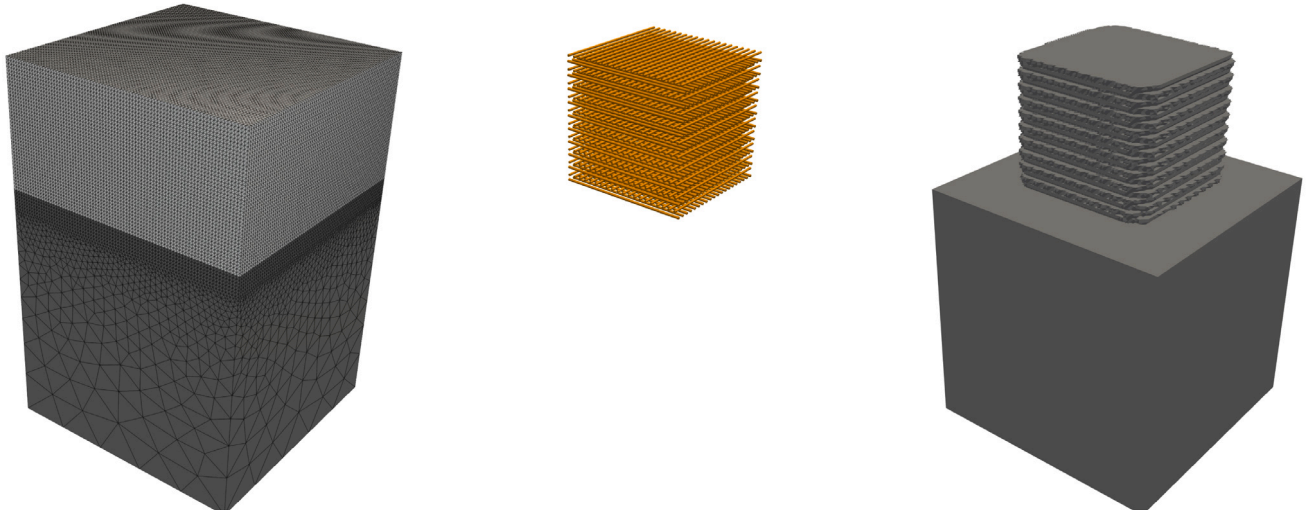
with a net laser power $P = 70.0$ W, radius $R = 100.0$ µm and speed $V = 1000.0$ mm s^{-1}.

The mesh, printing path and bulk material at the final time are shown in Figs. 20(a) to 20(c). The substrate is 2 mm thick, and 0.5 mm wider than the cube in the in-plane directions. A tetreahedral mesh is employed, with fine elements of size $\frac{R}{4}$ in the powder bed and coarse elements of size $4R$ at the bottom of the substrate. The scanning path is aligned with the axes and alternates direction from a layer to the next, with 10.0 s of cooling time in-between. The layer thickness and hatch spacing are both set to R. The fast subdomain is defined by padding the heating track by $3.5R$ in all directions. Fig. 20(c) shows an example of the resulting bulk mesh after all layers have been printed and cooled.

The models compared are the Robin and Dirichlet substeppers without an advected subdomain, as well as the Robin substepper with an advected subdomain. For the advected subdomain case, both monolithic and staggered treatments of Eq. (10) are considered. All speed-ups are reported relative to the unsubstepped standard model. Throughout this section, the fast time-step Δt_f is fixed at $0.5T_{hs}$ for the models without an advected subdomain, whereas in the advected subdomain case it is progressively increased as the melt pool approaches steady state, following the criterion in inequality (11).

Table 6 reports the speed-ups obtained by the various substeppers for increasing cube sizes. The ratio of fast DOFs to total DOFs is given to contextualize the observed gains. Timings are collected in serial in order to factor out the effects of load imbalances, and prints are truncated to a single layer to make simulation times manageable.

Across all domain sizes considered, the Robin substepper consistently outperforms the Dirichlet substepper. This is expected, as the Dirichlet approach requires solving the joint system (16) at every substepping cycle, whereas the Robin substepper solves the two subdomains sequentially, leveraging the locality of the fast dynamics. For instance, in the 2 mm cube case, the assembly and solution of the joint system account for 38% of the total assembly-solve time for $\Delta t_s = 5T_{hs}$, and 20% for $\Delta t_s = 10T_{hs}$.

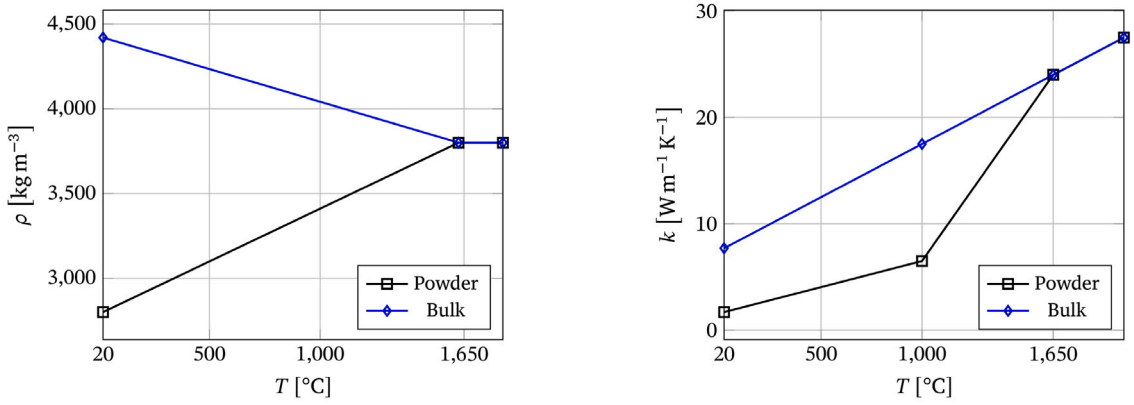


(a) Mesh and material composition at start of print. ( ) Powder ( ) Bulk.

(b) Printing path.

(c) Bulk material at final time.

Fig. 20. 1.0 mm cube. Powder elements corresponding to each layer are activated during recoating time-steps.



(a) Density

(b) Conductivity

Fig. 21. Thermally dependent material parameters for Ti-6Al-4V powder and bulk.

Table 6
Speed-up factors relative to the unsubstepped model for increasing edge lengths of the cubic domain. The models without an Advected Subdomain (AS) employ $\Delta t_f = 0.5T_{hs}$, while the models with an AS progressively increase their Δt_f. The maximum Δt_f reached ranges from $2T_{hs}$ for the 1mm cube to $3T_{hs}$ for the 3.0 mm cube.

Edge length [mm]	Δt_s [T_{hs}]	Fast DOF [%]	Speed-up			
			Robin No AS	Dirichlet No AS	Robin Monolithic AS	Robin Staggered AS
1.0	5.0	22.0	2.0	1.7	1.9	1.4
	10.0	31.0	1.8	1.7	1.8	1.3
1.5	5.0	14.0	2.7	2.5	2.7	2.2
	7.5	17.0	2.7	2.5	2.9	2.2
2.0	5.0	10.0	3.3	3.0	3.6	3.3
	10.0	14.0	3.3	3.1	4.4	3.2
2.5	5.0	7.4	3.8	3.4	4.2	3.5
	12.5	11.9	3.9	3.7	5.4	4.1
3.0	5.0	5.5	4.2	3.7	4.7	4.0
	7.5	6.6	4.7	4.2	5.7	4.7
	10.0	7.7	4.8	4.4	6.4	5.1

Table 7
Fractional breakdown of computational costs for the 2.0mm cube cases reported in Table 6. Columns indicate the percentage of joint assembly–solve time spent in residual assembly (Res.), Jacobian assembly (Jac.), and the linear solve (LS). All solvers employ GMRES with BoomerAMG preconditioning from the HYPRE library and the default Newton–Raphson settings of PETSc's SNES.

Substepper	Advected subdomain	$\Delta t_s = 5T_{hs}$			$\Delta t_s = 10T_{hs}$		
		Res.	Jac.	LS	Res.	Jac.	LS
–	No	45%	33%	22%	–	–	–
Robin	No	30%	45%	25%	29%	51%	20%
Dirichlet	No	31%	43%	26%	29%	51%	20%
Robin	Yes, monolithic	34%	41%	25%	35%	47%	18%
Robin	Yes, staggered	31%	41%	27%	30%	46%	24%

Table 7 reports additional details on the fractional breakdown of the computational costs. Assembly costs dominate the overall computational time, which explains why the staggered treatment of the advected subdomain does not lead to a significant cost reduction compared with the monolithic approach: even though the staggered approach isolates the asymmetric part of the Jacobian, the linear solve is only about a fifth of the total cost.

The staggered approach typically requires three to four iterations to converge each micro-step when using the Robin coefficients proposed in [20]. Nonlinear convergence is generally more challenging for the monolithic approach, so the staggered formulation can occasionally be more efficient, for example, at the beginning of the print when the material is still cold. However, as the melt pool stabilizes nonlinear convergence becomes easier and the advantage of the staggered method diminishes.

When evaluating the effect of the advected subdomain on performance, it is found that its benefit increases with track length, yielding substantial speed-ups for cubes sizes over 2.0 mm. This behavior is expected: as the melt pool has more time to stabilize, the fast time-step can grow further.

Note from Table 6 that the Dirichlet substepper tends to respond better to increases in Δt_s, since it decreases the number of joint solves required. Interestingly, for the 1.0 mm cube, all substeppers perform worse at $\Delta t_s = 10T_{hs}$ than at $\Delta t_s = 5T_{hs}$. As Δt_s increases, the communication overhead between the partitions decreases, but the fraction of DOFs in the fast subdomain also increases. Hence, for a Δt_s too large relative to the domain size, the fast DOF ratio may exceed the profitability threshold; Hodge [1] suggests a value around 30%.

After assessing the performance of the various substepping schemes, their solution fields are now compared. Fig. 22 shows material composition and temperature contours at an intermediate time for the 1.0 mm cube with $\Delta t_s = 10T_{hs}$. The elements of the fast and advected subdomains, Ω_f (tetrahedra) and Ω_m (hexahedra), are highlighted in Figs. 22(b) and 22(c). Since the macro-step spans an entire track, the fast subdomain extends into the powder bed and does not intersect the thermal tail.

It is observed that, in the absence of an advected subdomain, the Robin and unsubstepped solutions are very similar. However, the presence of the advected subdomain leads to a visibly different thermal profile, with a thermal tail aligned with the scan direction.

This behavior occurs because the advected subdomain overlaps with the thermal tail left by the previous track: the interhatch spacing is $1R$, while the advected subdomain width is $1.5R$. The former is quite common in practice, and the latter is hard to further reduce without excluding part of the heat source from the advected subdomain. Consequently, the advection shears away the previous thermal tail, resulting in a profile aligned with the current track, and ultimately underestimating the heat diffusion into the previously printed material.

This is corroborated by Fig. 23. The left panel shows the temperature history at a probe, and the right panel the melt pool width throughout the print. The latter further quantifies the aforedescribed shearing effect, while the former shows that thermal diffusion in the depth direction is also affected by advection.

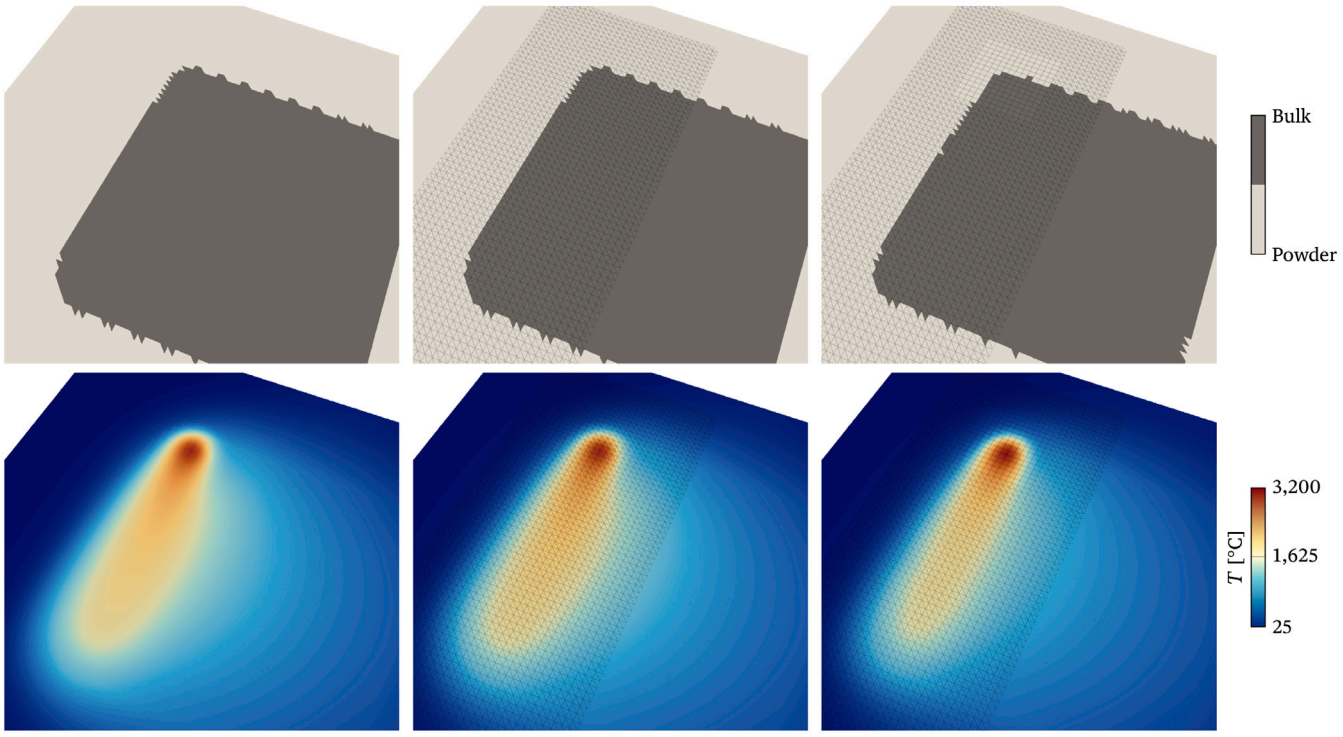
The other substeppers otherwise agree quite well with the unsubstepped solution. As shown by the temperature history, the Dirichlet substepper provides a slightly more accurate prediction than the Robin substepper in this case.

5. Conclusions

The current work presents a comprehensive study on substepping methods for LPBF modeling, and more generally, moving heat source problems with large contrasts in time scales.

- The proposed Robin–Robin scheme is shown to be relatively cheap in comparison to other methods in the literature, while being as or more accurate. More generally, a joint fast-slow solve in the substepping cycle does not appear to be required for accuracy, provided that the underlying domain decomposition scheme is well chosen.
- The Dirichlet substepper benefits from alternative predictor strategies, such as a purely diffusive step without the heat source. The latter is shown to be more accurate, especially for large Δt_s, and results in easier nonlinear solves during both the predictor and micro-steps; the original predictor can lead to unphysical temperature spikes at the fast-slow interface which grow worse with Δt_s.
  A possible direction for future work is the development of a Robin variant of the semi-monolithic scheme. Such an approach would be less sensitive to the predictor choice than the Dirichlet scheme and could achieve higher overall accuracy than either of the schemes presented in this work.
- Results of the first numerical example indicate that the current bottleneck for larger macro-steps is the time integration scheme of the slow problem. Most references in the literature hover around similar macro-step sizes of $5-10T_{hs}$. The example also shows that the truncation of the substepping loop to a single iteration does not significantly impact accuracy.
- The underlying model is shown to be relevant, as it is able to reproduce experimental melt pool geometries. However, it required the use of an unphysical dip in absorptivity to do so, suggesting that a heat sink mechanism is missing from the model. Said mechanism could be related to evaporation or fluid flow.
- The advected subdomain approach can yield significant speed-ups, as demonstrated in the last numerical example, but these benefits are largely limited to cases with long tracks and few direction changes. Moreoever, the model produces relatively large errors when a track overlaps with the thermal tail of previous tracks, reducing its general applicability. The moving mesh and advection terms add considerable complexity to both implementation and solver strategies. This approach is therefore best suited when high accuracy in the HAZ is required and the scan pattern is simple, rather than as a general-purpose speed-up technique.
  To improve the advected subdomain approach, future work could investigate spatially and temporally varying advection velocities. The former can be achieved with an Arbitrary Lagrangian–Eulerian framework, and would allow the advected region to shrink dynamically and minimize its influence. The latter could be achieved by gradually ramping up the advection from zero to the laser speed at the start of a track, thereby smoothing the transition between the non-advected and advected regimes. This would likely mitigate the temperature spike observed when the switch occurs within a single time-step.

Taken together, these results show that substepping schemes offer a computationally efficient path for LPBF thermal simulations without a substantial loss of accuracy. Future work should investigate



(a) Unsubstepped. (b) Robin. (c) Robin with advected subdomain.

Fig. 22. Temperature contour and material composition at intermediate time for the 1 mm cube case with $\Delta t_s = 10T_{hs}$. Tetrahedral elements correspond to the fast subdomain, and hexahedral elements to the advected subdomain.

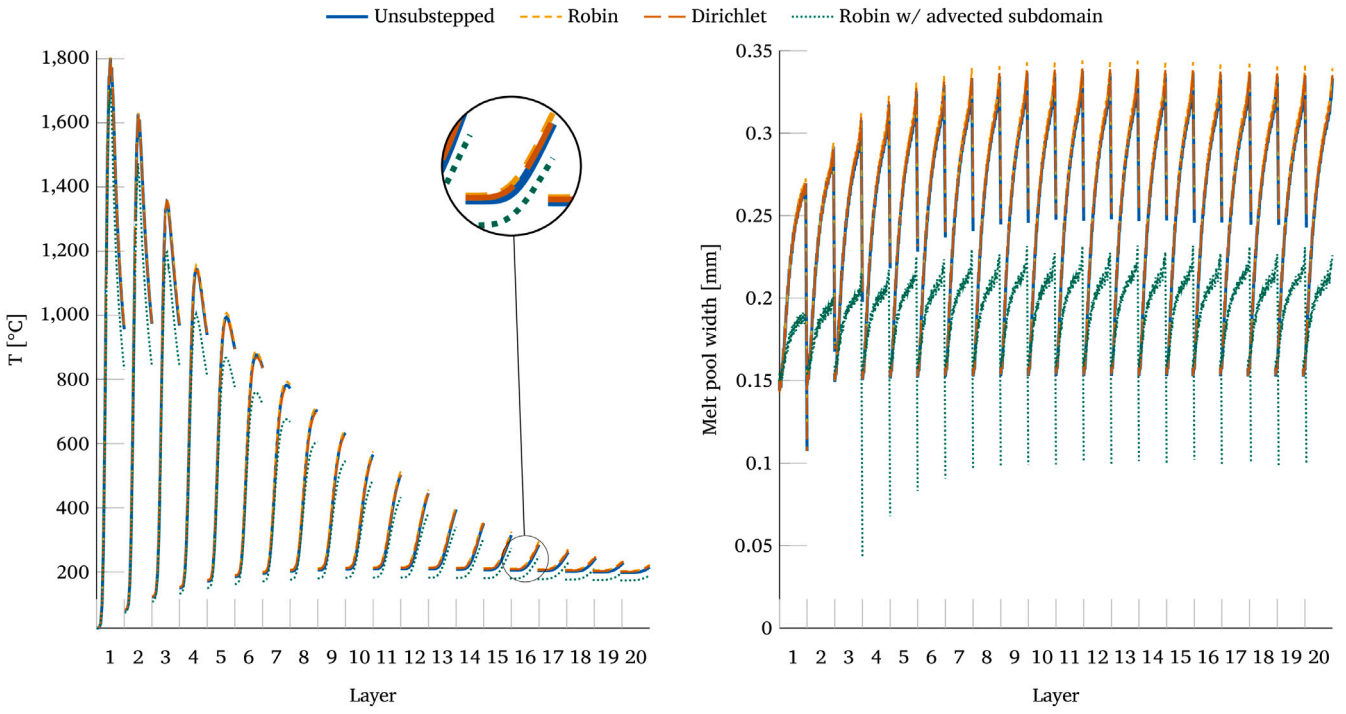


Fig. 23. Temperature history at a point (left) and melt pool width (right) throughout the print of the 1.0 mm cube using $\Delta t_s = 10T_{hs}$. The probe is located at the center of the top surface of the substrate, that is, right below the first layer. Cooling intervals are excluded for clarity.

thermo-mechanical extensions, improved physical models for heat loss mechanisms, and improved time integration schemes for the slow problem to enable larger macro-steps.

CRediT authorship contribution statement

Mehdi Slimani: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Michele Chiumenti:** Writing – review & editing, Supervision, Project administration. **Miguel Cervera:** Supervision.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, Mehdi Slimani used ChatGPT (OpenAI) in order to improve the readability and clarity of the text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Michele Chiumenti gratefully acknowledges financial support from the Spanish Ministry of Science, Innovation and Universities through the R+D project "*Desarrollo de un gemelo digital para monitorización y el control activo del proceso de fabricación aditiva DED (DIGITAM)*" (Ref. CPP2023-010440), as well as from the Center for Technological Development and Innovation (CDTI) through the R+D project "*Machine Learning Driven Metal Additive Manufacturing By Direct Energy Deposition For Repairing Operations (MALE4RAM)*".

Mehdi Slimani is grateful to the FEniCSx project and its community for providing an actively developed open-source FEM platform and for their consistently helpful and responsive support.

Appendix A. Weak forms of advected subdomain model

This appendix provides the weak forms of the moving and fixed subproblems of the advected subdomain approach. The latter is presented first, as it is comparatively simpler. After spatial discretization using FEM and temporal discretization using the Backward-Euler scheme, the residual form of Eq. (6) with boundary and interface conditions is given by

$$\begin{aligned}\mathcal{R}_f = &\int_{\Omega_f} \phi_f \rho c_p \frac{T_f^{n+1} - T_f^n}{\Delta t} + \int_{\Omega_f} \phi_f \rho L \frac{f_l(T_f^{n+1}) - f_l(T_f^n)}{\Delta t} \\ &+ \int_{\Omega_f} \nabla\phi_f \cdot \left(k \nabla T_f^{n+1}\right) \\ &- \int_{\partial\Omega_f \setminus \Gamma} \phi_f \left(h_{\text{conv}} \left(T_f^{n+1} - T_{\text{env}}\right) + \varepsilon\sigma \left(T_f^{n+1^4} - T_{\text{env}}^4\right)\right) \\ &- \int_{\Gamma} \phi_f \left(k\Big|_{T_m^{n+1}} \partial_n T_m + (T_m^{n+1} - T_f^{n+1})\right) = 0 \qquad \forall \phi_f \in V_f\end{aligned} \tag{A.1}$$

Conversely, the treatment of the moving subproblem is more involved, as the advection terms in Eq. (5) require stabilization. In this work, *spatial-only* VMS (Variational MultiScale) stabilization is applied, as described in [34]. To derive the corresponding terms, the PDE (5) is first rewritten in terms of a strongform residual R and a spatial differential operator $\mathcal{L}$:

$$R_T T = 0 \tag{A.2}$$

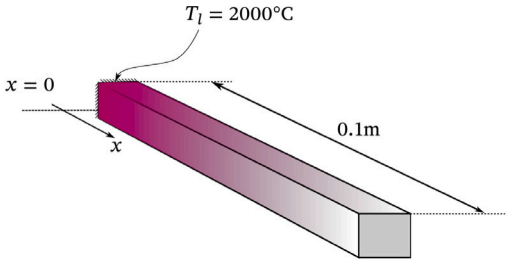


Fig. B.24. Schematic of the one-dimensional melting bar model with a Dirichlet boundary condition at $x = 0$.

$$R_w u := \rho c_p \partial_t u + \rho L \partial_t f_l\Big|_u + \mathcal{L}_w u - r$$
$$\mathcal{L}_w u := -(\rho c_p + \rho L f_l'\Big|_w)\mathbf{v} \cdot \nabla_\xi u - k\Delta_\xi u$$

VMS stabilization, like the Streamline Upwind Petrov–Galerkin method, adds residual-based diffusion in the advection direction to the weak form. The residual (A.2) is weighted by the test function, an elementwise stabilization parameter τ, and the adjoint operator $\mathcal{L}^*$, which reverses the sign of the advection term:

$$\begin{aligned}\mathcal{R}_m = &\int_{\Omega_m} \phi_m \rho c_p \frac{T_m^{n+1} - T_m^n}{\Delta t} + \int_{\Omega_m} \phi_m \rho L \frac{f_l(T_m^{n+1}) - f_l(T_m^n)}{\Delta t} \\ &- \int_{\Omega_m} \phi_m \left(\rho c_p + \rho L f_l'(T_m^{n+1})\right) \mathbf{v} \cdot \nabla T_m^{n+1} + \int_{\Omega_m} \nabla\phi_m \cdot \left(k \nabla T_m^{n+1}\right) \\ &- \int_{\partial\Omega_m \setminus \Gamma} \phi_m \left(h_{\text{conv}} \left(T_m^{n+1} - T_{\text{env}}\right) + \varepsilon\sigma \left(T_m^{n+1^4} - T_{\text{env}}^4\right)\right) \\ &- \int_{\Gamma} \phi_m \left(k\Big|_{T_f^{n+1}} \partial_n T_f + (T_f - T_m^{n+1})\right) \\ &- \int_{\Omega_m} \left(\mathcal{L}^*_{T_m^{n+1}} \phi_m\right) \tau R_{T_m^{n+1}} T_m^{n+1} - \int_{\Omega_m} \phi_m r = 0 \qquad \forall \phi_m \in V_m \\ \tau = &\left(\frac{4k}{h^2} + \frac{2\|\mathbf{v}\|\rho(c_p + L f_l'(T_m^{n+1}))}{h}\right)^{-1}\end{aligned} \tag{A.3}$$

$$R_T T = \rho c_p \partial_t T + \rho L \partial_t f_l\Big|_T - (\rho c_p + \rho L f_l'(T_m^{n+1}))\mathbf{v} \cdot \nabla_\xi T - k\Delta_\xi T - r$$
$$\mathcal{L}^*_T v = +(\rho c_p + \rho L f_l'(T_m^{n+1}))\mathbf{v} \cdot \nabla_\xi v - k\Delta_\xi v \tag{A.4}$$

Appendix B. Comparison of apparent heat capacity and source-based latent heat treatments

This section compares the performance of the apparent specific heat capacity (c_{app}) model and the source-based model introduced by Kollmannsberger et al. [21] and Celentano et al. [22], focusing on nonlinear convergence and L2 error.

For this study, the one-dimensional melting bar problem proposed by [21] is revisited. The setup consists of a semi-infinite rod, initially solid and at a uniform temperature T_s. At $t = 0$, the rod is suddenly heated at $x = 0$ with a Dirichlet boundary condition $T(0, t) = T_l$. The bar then undergoes isothermal melting with phase change occurring at the melting temperature $T_m < T_l$. The problem is illustrated in Fig. B.24.

The analytical solution was originally derived by Neumann, and is described in [21,35]. The solid–liquid interface $X(t)$ advances in the positive x direction according to

$$X(t) = 2\lambda\sqrt{\alpha t}$$

where α is the thermal diffusivity and λ is the solution of the transcendental equation

$$\frac{St}{\exp\lambda^2 \operatorname{erf}\lambda} - \frac{St\sqrt{\alpha}}{\sqrt{\alpha}\exp\lambda^2 \operatorname{erfc}\lambda} = \lambda\sqrt{\pi} \qquad St = \frac{c_p\left(T_l - T_m\right)}{L} \tag{B.1}$$

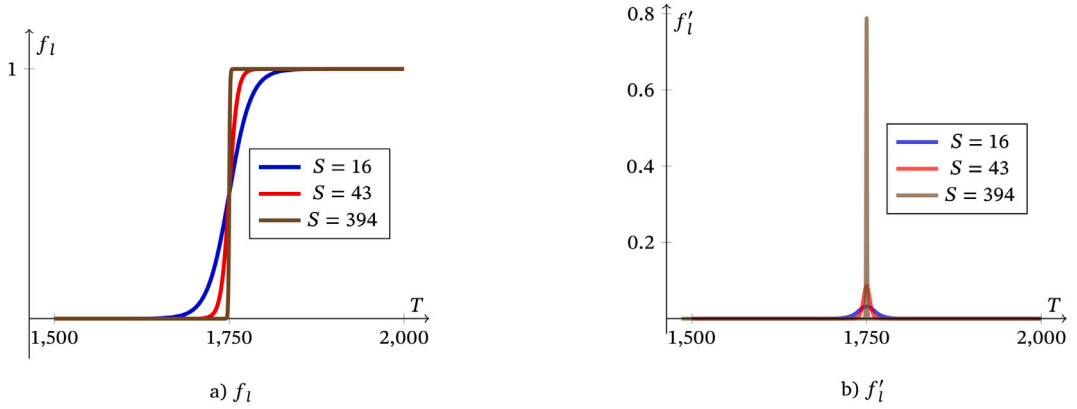


Fig. B.25. Liquid fraction f_l and its derivative f_l' for various S values.

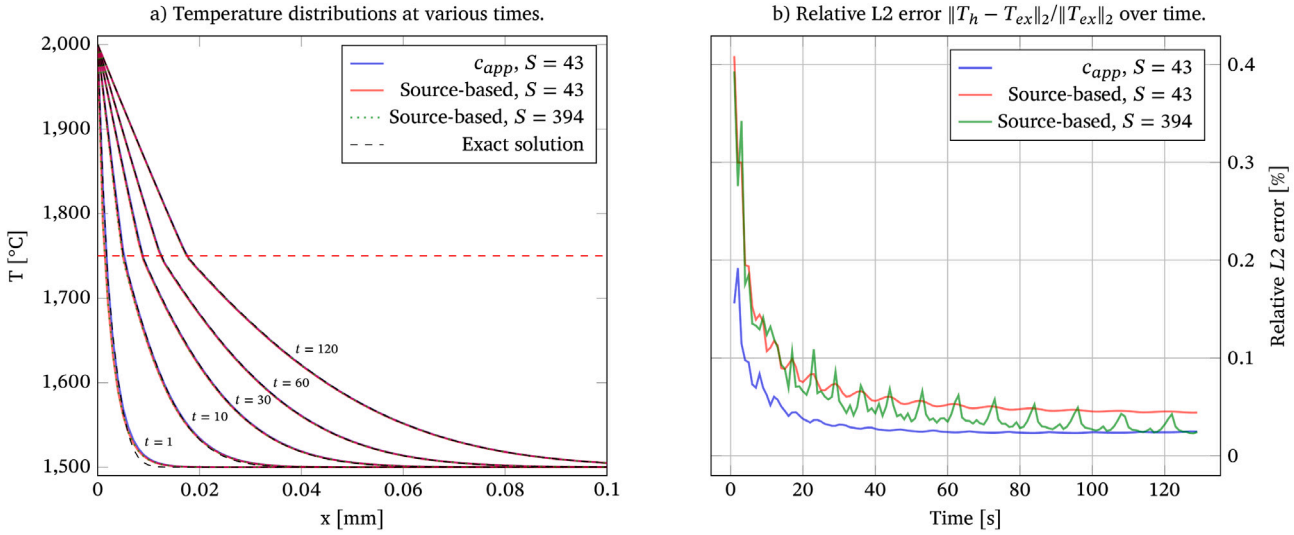


Fig. B.26. Temperature distributions and comparison of relative $L2$ error for $h = 0.01$ m and $\Delta t = 1$ s. The apparent heat capacity and source-based models reach $S_1 = 43$ and $S_2 = 394$, respectively. Errors obtained by the apparent specific heat capacity model are slightly lower.

Table B.8
Material parameters for the melting bar problem.
Source: Values are taken from [21].

Parameter	Value	Unit	Parameter	Value	Unit
ρ	4510.0	$\mathrm{kg\,m^{-3}}$	T_s	1500	°C
c_p	520.0	$\mathrm{J\,kg^{-1}\,K^{-1}}$	T_l	2000	°C
k	16.0	$\mathrm{W\,m^{-1}\,K^{-1}}$	T_m	1750	°C
			L	325.0	$\mathrm{kJ\,kg^{-1}}$

where St is the Stefan number. The analytical solution is then given by

$$T_{ex}(x,t) = \begin{cases} T_l - (T_l - T_m)\dfrac{\operatorname{erf}\left(x/2\sqrt{\alpha t}\right)}{2} & x \le X(t) \\ T_s + (T_m - T_s)\dfrac{\operatorname{erfc}\left(x/2\sqrt{\alpha t}\right)}{\lambda} & x > X(t) \end{cases}$$

where erf and erfc are the error function and its complement, respectively. Material parameters resembling Ti-6Al-4V are used, as listed in Table B.8. A value $\lambda = 0.30568105$ is computed numerically by applying a root-finding algorithm to Eq. (B.1) with the given parameters.

Eq. (3) is solved using both the apparent specific heat capacity and the source-based methods to treat the latent heat terms. Note that, while the original problem involves strictly isothermal phase change, the regularized liquid fraction f_l introduces a narrow mushy region between T_s and T_l, which shrinks as S increases. Hence, the robustness of both models is assessed by progressively increasing S until the nonlinear solvers cease to converge. Fig. B.25 shows f_l and its derivative for representative values of S considered in this section.

The problem is marched in time from $t = 0$ to $t = 120$ s using the Backward-Euler method with $\Delta t = 1$ s. A uniform mesh with linear elements of size $h = 0.01$ mm is employed, and a high order quadrature rule (degree 4) is used to capture the subgrid variations in the regularized liquid fraction f_l. The implementation uses `FEniCSx` [27] with the default `SNES` nonlinear solver from `PETSc` to solve the Newton–Raphson problems.

The authors find that the apparent specific heat capacity model fails to converge beyond $S_1 = 43$, whereas the source-based model reaches $S_2 = 394$. As seen in Fig. B.25, it is clear that S_2 is practically indistinguishable from the isothermal phase change limit.

Fig. B.26(a) compares the temperature profiles predicted by both models at various times. Small differences are visible at $t = 1$ s and $t = 10$ s, but by $t = 30$ s both solutions are in good agreement with the analytical result. Interestingly, despite the lower S value, the apparent specific heat capacity model achieves slightly lower L2 error, as shown in Fig. B.26(b).

However, the picture changes with finer discretizations. After refining the mesh size by a factor of 10, the apparent specific heat capacity

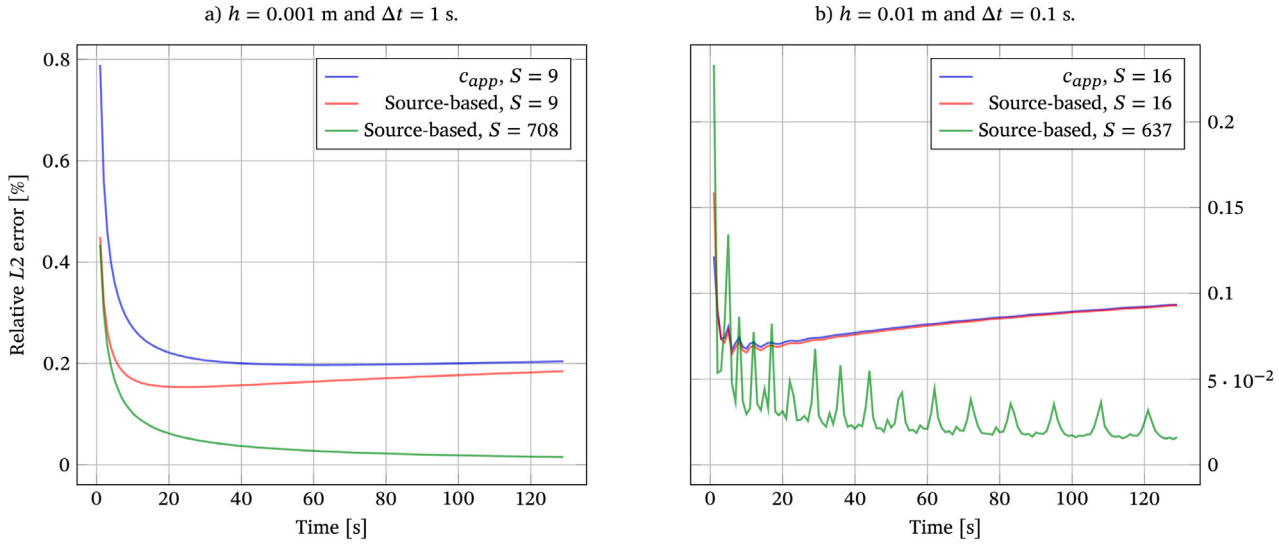


Fig. B.27. Relative $L2$ errors over time for refined mesh size (left) and refined time-step (right). The source-based models reaches much higher S values and achieves lower errors.

and source-based models are able to converge $S_1 = 9$ and $S_2 = 708$, respectively. Likewise, after refining the time-step by a factor of 10 while keeping the original mesh size, they converge up to $S_1 = 16$ and $S_2 = 637$. Moreover, the source-based model achieves lower errors in both cases, as shown by Fig. B.27, with the difference becoming more pronounced for the larger S values.

In conclusion, it is difficult to declare a clear winner in terms of accuracy: for the base discretization, the c_{app} model fares better, whereas for finer discretizations the source-based model is preferred.

Nevertheless, the source-based model is clearly more robust, as it can approximate the isothermal phase change limit much more closely. Based on the authors' experience, the apparent specific heat capacity model often fails to converge for $S > 1$ when applied to LPBF problems. For this reason, the source-based model is employed in this work. The authors further note that the adoption of an apparent heat capacity formulation may explain why some references [1,16,18] report the need for very fine time-steps ($\Delta t < 0.1 T_{hs}$).

Data availability

No data was used for the research described in the article.

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